National University of Science and Technology

Think in Other Terms

FACULTY OF APPLIED SCIENCE

DEPARTMENT OF APPLIED BIOLOGY AND BIOCHEMISTRY

BIOTECHNOLOGY

A MOLECULAR SURVEY OF ANTIFUNGAL SUSCEPTIBILITY PATTERNS AND A SELECTED VIRULENCE FACTOR OF *CANDIDA ALBICANS* FROM MZINGWANE AND MTSHABEZI DAMS

MPOFU SIMBARASHE MTHANDAZO: N02017758M

JUNE 2024

A dissertation submitted in partial fulfilment of the requirements for the Bachelor of Science Honours Degree in Biotechnology, within the department of Applied Biology and Biochemistry, Faculty of Applied Sciences at the National University of Science and Technology (NUST), 2024

FACULTY OF APPLIED SCIENCE

DEPARTMENT OF APPLIED BIOLOGY AND BIOCHEMISTRY

BIOTECHNOLOGY

A MOLECULAR SURVEY OF ANTIFUNGAL SUSCEPTIBILITY PATTERNS AND A SELECTED VIRULENCE FACTOR OF *CANDIDA ALBICANS* FROM MZINGWANE AND MTSHABEZI DAMS.

MPOFU SIMBARASHE MTHANDAZO

JUNE 2024

SUPERVISOR: DR. E. ZUMBIKA

ABSTRACT

This project was carried out to determine the antifungal susceptibility patterns and detect antifungal resistance genes (ERG11 and MDR1) and a virulence gene (ALS1) of *Candida albicans* isolated from environmental water samples. The samples were collected from Mzingwane and Mtshabezi Dams. The samples were collected from different sites of the dams to make composite samples so as to get an average representation of the sampling locations. The water samples were enriched in nutrient broth then subcultured on nutrient agar, Sabouraud dextrose agar (SDA) and Candida Ident Agar respectively to get pure isolates of *Candida albicans*. The isolates were confirmed to be *Candida albicans* using the germ tube test, lactophenol cotton blue assay and a chromogenic agar assay using Candida Ident Agar. A total of 20 isolates were collected from the environmental water samples. Antimicrobial susceptibility tests were done using fluconazole, itriconazole and posaconazole. Of the 20 isolates collected, all 20 (100%) isolates were resistant to fluconazole, 17 (85%) were resistant to itriconazole and 10 (50%) were resistant to posaconazole. DNA extraction was done and conventional PCR to amplify ERG11, MDR1 and ALS1 genes. For ERG11, 9 isolates were selected for gene amplification and all 9 (100%) were positive for ERG11: for MDR1, 9 isolates were selected for gene amplification and 7 (77,8%) isolates were positive for MDR1 and for ALS1, 6 isolates were selected for gene amplification and 5 (83,3%) were positive for ALS1. Of the 4 isolates that were positive for ERG11, MDR1 and ALS1, all were resistant ot fluconazole, itriconazole and posaconazole.

DECLARATION

I, Mpofu Simbarashe Mthandazo, hereby declare that the project entitled "A Molecular Survey of Antimicrobial Susceptibility Patterns and a Selected Virulence Factor of *Candida albicans* from Mzingwane Dam and Mtshabezi Dam" submitted to the National University of Science and Technology in partial fulfilment of the Bachelor of Science Honours Degree in Biotechnology is a record of bona fide project work carried out by myself under the guidance of Dr E. Zumbika. I further declare that the work reported in this project has not been submitted and will not be submitted either in part or in full, for the award of any other degree in this University or any other institute.

Signed: ____________________

Date: _____/______/______

ACKNOWLEDGEMENTS

It is with honour that I extend my absolute gratitude to several people who contributed to the success of my Bachelor of Sciences Honours in Biotechnology thesis. In particular, I extend my profound gratitude to my project supervisor, Dr E. Zumbika, for his guidance and knowledge he offered towards the completion of my project. I would also like to thank the assistance of Dr. N. Nleya for the molecular biochemicals and of Mrs. M. Sibula, Mrs. Maunganidze, Miss Sthembile Sihambi and Mr. Jefferson Metha for knowledge and technical assistance they offered during the course of this research. I extend also my gratitude to my colleagues who worked with me and contributed their time, knowledge and enthusiasm towards the betterment of my project. I am also grateful to my family for their support, financial, emotional or otherwise, and insights and words of encouragement they offered towards the completion of my project. Lastly, I would like to thank the National University of Science and Technology for nurturing and sharpening my potential as a competitive student and also for the opportunity to learn, grow and mature as student.

DEDICATION

I dedicate this work to my parents and my friends who have been a constant source of support and encouragement during the challenges of graduate school and life. Finally, I dedicate this to God who has brought me through from the beginning of graduate school life till the end.

LIST OF TABLES

LIST OF FIGURES

LIST OF ABBREVIATIONS

PRR…………… Pattern Recognition Receptor

TLR…………… Troll-like Receptor

MR……………. Mannose Receptor

WHO…………... World Health Organisation

DNA…………… Deoxyribonucleic Acid

PCR……………. Polymerase Chain Reaction

VVC……………. Vulvovaginal Candidiasis

IC………………… Invasive Candidiasis

AIDS…………… Acquired Immunodeficiency Syndrome

HIV……………... Human Immunodeficiency Virus

MDR……………. Multi Drug Resistance

OC………………... Oral Candidiasis

IBD………………. Inflammatory Bowel Disease

AMP……………… Antimicrobial Peptides

ALS………………. Agglutinin-like Sequence

MRR……………… Multidrug Resistance Regulator

TAC………………. Trasncriptional Activator of CDR

CDR………………. Candida Drug Resistance

HWP………………. Hyphal Wall Protein

ROS……………...... Reactive Oxygen Species

RNS……………..... Reactive Nitrogen Species

MIC………………. Minimum Inhibitory Concentration

LOH………………. Loss of Heterozygosity

MTL………………. Mating-type Locus

CLSI……………… Clinical and Laboratory Standards Institute

Table of Contents

CHAPTER 1

INTRODUCTION

Candida albicans is a dimorphic yeast fungus that is part of the *Candida* genus that consists about 200 to 300 species of which 20 of those are known to be pathogenic, amongst which are *Candida auris*, *Candida tropicalis*, *Candida glabrata*, and *Candida albicans*. *Candida* species are commensals and thus are part of the normal human flora and are localized on skin and gastrointestinal and genital tracts. However, *Candida* species can also cause various infections in susceptible patients that includes the elderly, hospitalized, or immunosuppressed patients. Amongst the different *Candida* sp., *Candida albicans* is the most commonly recovered (37%) from clinical species, followed by *Candida glabrata* (27%). Other clinically relevant species recovered from blood stream infections include *C. parapsilosis* (14%), *C. krusei* (2%), *C. tropicalis* (8%), *C. dubliniensis* (2%), *C. lusitaniae* (2%), and the most recent, *C. auris*. *C. auris* is an emergent multi-drug-resistant pathogen that is often misidentified and at present is a major concern in healthcare settings. *C. auris* reported cases increased by 318% between 2015 and 2018 (Bhattacharya *et al*., 2020).

Over the past few years, the increase in the incidence of fungal infections in several countries has stimulated the scientific community to pay closer attention to this problem. (Banerjje *et al*., 1991; Anaissie, 1992; Fox, 1993) At the time of publishing the articles, Banerjje and co-workers (1991), Anaissie (1992) and Fox (1993) had stated that fungal infections are not as important – using quantitative parameters – as bacterial ones, and presumably their incidence will slowly decrease as HIV infection is gradually controlled in several countries. Unfortunately, current trends of fungal infections show that occurrence of fungal infections have increased drastically. However, their treatment is still complicated for systemic infections – the most severe ones – due to the close functional similarity of the micro-organism and the host mammalian cells. Currently, available therapies rely mainly on the polyene and azole compounds. Amphotericin B, the gold standard in the treatment of fungal infections, presents serious side effects – now partially overcome with the development of new ways of administration such as liposomal formulations – that restrict its use to hospitals (Dupont, 1992; Bennett, 1996). On the other hand, the emergence of resistance to azoles as a consequence of long-term therapy may limit their usefulness in the future (Rex *et al*., 1995; Sanglard *et al*., 1995; van den Bossche, 1998).

Although in recent years there has been an increase in the diversity of pathogenic fungi isolated from clinical samples, *Candida albicans* is still the predominant cause of fungal infections. This yeast is a member of the commensal flora, and in a certain percentage of individuals it is found in their gastro-intestinal and vaginal tracts (Odds, 1988; Odds, 1994). *C. albicans* is normally considered to be an opportunistic pathogen, since in situations that cause a decrease in host defences; it is able to access different locations of the human body and cause disease. One of the main characteristics of this pathogenic yeast is its ability to switch from a unicellular to a hyphal mode of growth, a property called dimorphism. Dimorphism is triggered in response to certain environmental conditions, such as temperature, pH or serum availability, and it is thought that it is this change which allows the organism to invade tissues and, therefore, contributes to the virulence of this micro-organism (Ryley and Ryley, 1990; Corner and Magee, 1997; Kobayashi and Cutler, 1998; Magee, 1998). All these reasons can be invoked to account for the importance of *C. albicans* as a model system to study fungal dimorphism and virulence as well as to identify new antifungal targets. Although the definition of a virulence factor may be difficult to achieve for a commensal organism, the identification of such functions – if properly substantiated by experimentation – could be of primary importance for the development of novel therapies.

Candida albicans is a diploid parasite that as often as possible causes mucosal and fundamental contaminations in people (Vincent *et al*., 2009). *Candida* species can colonise a few particular anatomical locales. The greater part of infections caused by commensal microorganisms comes from endogenous colonisation. Exogenous pollution, for example, diseases communicated through emergency clinic workers, medical clinic air and biofilm-debased intrusive gadgets like catheters, can also occur (Ingham *et al*. 2012; Correia *et al*.; 2015; Limon *et al*.; 2017). Diseases brought about by *Candida albicans* can be delegated shallow, cutaneous, mucosal, and fundamental infection. At the point when *Candida* spp. taint the oral cavity, skin, genitalia, respiratory framework, and the remainder of the gastrointestinal lot, the disease is delegated the shallow sort. Intrusive candidiasis is a disease portrayed with very extreme conditions, for example, candidemia, meningitis and endocarditis (De Rosa *et al*., 2009). In hospitalised patients and those with bedraggled safe framework, intrusive contamination is a huge reason for dismalness and mortality along with increased frequency as well as pervasiveness rates.

The pathogenesis of *Candida* species is a complex cycle including numerous instruments and pathways. It is, in a similar manner, a multifactorial system, including highlights of both the host and the microorganism (Negri *et al*., 2010). For contamination to be set up, the pioneering microorganism should avoid immune system defences, duplicate in the host environment, and survive in the safe arrangement of the host. The living being must have the option to spread to other body tissues and organs, especially in foundational disease (Silva *et al*., 2011). If the skin and gastrointestinal boundaries are compromised, profound organ candidiasis is bound to occur. In significant instances, circulatory system intrusion may be possible which will disperse to various organs of the body. *Candida* contaminations in many people are asymptomatic. This is due to the host's immunological framework that bars the commensal organism from spreading in the body. In any case, a change in microbiota balance combined with different elements, can work with the spread of *Candida albicans* which is regularly deadly in 42% of announced cases (Wisplinghoff *et al*., 2003; Bongomin *et al*., 2017; Dadar *et al*., 2018). *C. albicans* is answerable for about half of candidiasis and non-*albicans Candida* species are liable for the rest of the *Candida* contamination. Diseases that are brought about by several other species of *Candida* are of extraordinary concern. A portion of these non-*albicans Candida* species are currently viewed as microbes with the potential to cause serious infections (Caceres *et al*., 2019). Preventing and reducing the incidences of *Candida* infections is an ongoing endeavour in human medicine. Even with the use of antifungal medications, scattered candidiasis is responsible for a high death rate, around 40–60% (Aslam *et al*., 2018). *Candida* is one of the main sources of mucosal contaminations. It additionally causes initial diseases particularly in immunosuppressive patients, regardless of its status as a commensal microorganism (Kornitzer *et al*., 2019). As anyone might expect, it is the pathogenic qualities and components that have gotten the most attention from specialists throughout the long term. As of late, much has been discovered about the components of *Candida albicans* pathogenesis. Studies have shown that at the core of the capacity of *Candida albicans* to multiply, change from non-destructiveness commensal to pioneering pathogenic organism and build up disease in the host lie in interconnected elements made out of transcriptional circuits, morphology-related/harmfulness encoding qualities, metabolic versatility, genome pliancy, phenotypic exchanging, biofilm arrangement, tissue harming extracellular hydrolytic catalysts, and a few different variables that work with destructiveness and pathogenesis in *Candida* species (Perez and Johnson, 2013). Changes in ecological pH, vigorous supplement procurement framework, escape from phagocytosis, avoidance from have insusceptible framework, the ability to have microbiome co-aggregation and protection from antifungals are other important characteristics that upgrade endurance and pathogenesis. In order to be capable of inducing such a broad spectrum of infections, *C. albicans* can live in several anatomically discrete sites and translates several virulence factors. The phenomenon of phenotypic converting from yeast- to filament-growth is a critical factor that contributes to the virulence of *C. albicans*. It offers a basis for activating different receptors leading to diverse immune responses. Other virulence factors of *C. albicans* contain adhesion factors, thigmotropism and secretion of several hydrolytic enzymes, such as lipase, phospholipase, and proteinase. In the past few years, it has become increasingly clear that PRRs (Pattern Recognition Receptors) are vital for the host response to *C. albicans*, with various TLRs (Troll-like Receptors) and Lectins having distinctive roles in innate immunity. Each ligand–receptor system activates specific intracellular signalling pathways, which in turn leads to modulation of various components of the host immune response. While a few receptors, like TLR4, dectin 1 and the MR (Mannose Receptors) apply more in this scenario, others employ immunosuppressive impacts (for instance, TLR2, CR3 and FcγR). After disclosure and characterised clarification of the part of TLRs in parasitic acknowledgment, further investigations have explained the job of the C-type lectin receptors with an emphasis on dectin-1 and dectin-2. The presence of various relationships among all of the components that guide the establishment of pollutions is an undeniable component in the pathogenesis of *Candida* species (Perez and Johnson, 2013).

1.1 Problem Statement

According to WHO (World Health Organisation), antimicrobial resistance is an emerging threat to global health and development. It requires worldwide cooperation to achieve sustainable development goals. Antimicrobial resistance is one of the top 10 global public health deteriorators. The abuse and overuse of antimicrobials are the prime factors behind developing drug-resistant pathogens. Antifungal resistance is an increasing problem with the fungus of the *Candida* genus. *Candida* infections may resist antifungal drugs, particularly the azole class of drugs. Azole drug-resistant *Candida* infections reduce treatment options and patients with candidemia are less likely to survive. Of the four classes of antifungal drugs, as the azole groups are used much, azole resistance is high (Dhasarathan *et al*., 2021). Prior to the introduction of antiretroviral therapy, there was an increase in the prevalence of fluconazole-resistant *C. albicans* among HIV-infected patients with oropharyngeal or esophageal candidiasis. Widespread use of itraconazole and fluconazole is thought to have been the major driver of azole resistance. Up to one-third of patients with advanced AIDS in one study harboured fluconazole-resistant *C. albicans* in their oral cavities. Azole-resistant *C.*

albicans is less common among patients with other diseases, such as vaginal candidiasis and candidemia. Other factors, such as exposure to antibacterial agents, immunosuppressive therapy, and the underlying medical condition of the host, might prove to be better predictors of the distribution of *Candida* species than fluconazole use (Kenafani and Perfect, 2024). Noting also that approximately 42% of all *Candida albicans* infections result in mortality, 75% of women have experienced VVC (vulvovaginal candidiasis) and 40−50% have had at least one extra episode of VVC and 5−8% of women suffer from recurrent VVC in their lives, *Candida albicans* is an opportunistic commensal pathogen and individuals that are immunocompromised have the highest risk of being infected (Sobel, 2007; Pappas *et al*., 2009).

1.2 Justification

Incorrect diagnosis for patients suffering from *Candida* infections occurs in two ways:

i with the use of empirical and pre-emptive strategies, treatment failure may relate to another diagnosis or multiple pathogens;

ii in patients receiving cytotoxic therapy, monoclonal antibodies, antiretroviral therapy, and other immune modulating therapies, the immune reconstitution inflammatory syndrome can dominate the clinical response to antifungal therapy. Because it is associated with prominent signs and symptoms of inflammation, immune reconstitution inflammatory syndrome can be confused with failure to control fungal growth (Kenafani and Perfect, 2024).

The emergence of drug resistance favours invasive candidiasis infections, leading to morbidity and mortality and leads to a global economic burden. With the advent of new resistance mechanisms, the efficiency of treating common infectious diseases may fail. The loss of microbial response to standard treatment will prolong the illness, increase hospital treatment expenditures, and give risk to the patient’s life. The abuse, misuse, and overuse of azole drugs for many common skin fungal ailments leads to *C. albicans* developing resistance and tolerance to the drug’s effect. A continuous drug pressure and the struggle for existence induce the threatened fungal species to develop molecular adaptation to resist the drugs, leading to the further spread of multi-drug resistance. The invasion of drug-resistant *C. albicans* promotes biofilm formation in catheters and other implants with drug-resistant microbial colonisation. The drug-resistant *C. albicans* pose a serious threat to immunocompromised cases, chemotherapy receivers, HIV patients, invasive candidiasis and also pulmonary mycosis patients. A greater global health issue may arise if pulmonary mycoses due to *C. albicans* and another opportunistic fungal pathogen with drug resistance is neglected. The epidemiology of the drug-resistant *C. albicans* has become a matter of great concern. The modifications in the molecular functions in *C. albicans* concerning the development of azole drug resistance need to be addressed (Dhasarathan *et al*., 2021)

1.3 Aim

To determine antifungal susceptibility patterns and detect antifungal resistance and virulence genes in azole resistant strains of *Candida albicans* from environmental specimens collected from Mzingwane and Mtshabezi Dams.

1.4 Objectives

- To isolate and obtain pure isolates of *Candida albicans* from water samples form Mzingwane and Mtshabezi Dams.
- To determine antifungal susceptibility patterns of *Candida albicans* isolates using the Kirby-Bauer disc diffusion assay.
- Perform a genotypic profiling of the antifungal resistance and virulence genes of *Candida albicans*

CHAPTER 2

LITERATURE REVIEW

2.1 *Candida albicans*: An Overview

Candida albicans appears in several morphological forms (blastospores, pseudohyphae, and hyphae). Blastospores divide asexually by budding (Molero *et al.*, 1998; Walker and White, 2017). During that process, new cell material is formed on the surface of the blastospore. The new bud grows from a small selected blastospore, and it is most often located distally from the site of a scar caused by birth, after which the phase of growth begins. After the growth phase ends, the cells divide, whereby the daughter separates from the parent cell by creating a partition (Molero *et al.*, 1998). Chains of elongated yeast cells characterise pseudohyphae, and the shape of hyphae is characterised by branched chains of tubular cells, with no narrowing at the sites of septation (Kornitzer, 2019). Filamentation is enhanced by a temperature higher than 37 °C, an alkali pH, serum, and high concentrations of CO_2 (Basso *et al.*, 2019). In the same way, it is also enhanced by a lack of nitrogen and carbon in the presence of N-acetylglucosamine (GlcNAc) (Kornitzer, 2019). This transition from a blastospore to a hypha is characterised by the activation of a complex regulatory network of signal paths, which include many transcription factors (Basso *et al.*, 2019). The main difference between yeast and hyphae composition is that the hypha wall has slightly more chitin content than yeast (Garcio-Rubio *et al.*, 2020). The cell wall is made of glucan, chitin, and protein. Its role is to protect the cell from stressful conditions in the environment, such as osmotic changes, dehydration, and temperature changes, and protect the cells from the host's immune defence (Höffs *et al.*, 2016; Reyna-Beltrán *et al.*, 2019). It is also responsible for adhesion to the host cell, with adhesion proteins such as Als1-7, Als9, and Hwp1 (Ciurea *et al.*, 2020). Communication of the cell with the outside environment takes place through the cell membrane (Hall, 2015). Sterols in the cell membrane are extremely important, giving the cell stability, rigidity, and resistance to physical stressors (Garcio-Rubio *et al.*, 2020). Ergosterol is the most represented sterol and is characteristic for the cell membrane of fungi. It is synthesized on the endoplasmic reticulum and lipid bodies (Jordá and Puig, 2020). In the cell membrane, there is a phospholipid bilayer containing proteins with the role of receptors, but also some whose role is transport and also signal transduction (Cho *et al.*, 2016). In its metabolism, *Candida albicans* uses glucose as a source of carbon and amino acids as nitrogen sources (Vianna *et al.*, 2020)

Virtually all humans are colonised with *Candida albicans*, but in some individuals this benign commensal organism becomes a serious, life-threatening pathogen. *C. albicans* possesses an arsenal of traits that promote its pathogenicity, including phenotypic switching (Alby and Bennett, 2009), yeast–hyphae transition (Kumamoto and Vinces, 2005) and the secretion of molecules that promote adhesion to abiotic surfaces (Chandra *et al.*, 2001). A balance is maintained between the ability of *C. albicans* to invade host tissues and the host's defence mechanisms (Kim and Sudbery, 2011; Kumamoto and Pierce, 2011). An alteration of this host–fungus balance can result in high levels of patient mortality (Pittet *et al.*, 1994; Charles *et al.*, 2003). Systemic *C. albicans* infections are fatal in 42% of cases (Wisplinghoff *et al.*, 2003), despite the use of antifungal therapies, and *C. albicans* is the fourth most common infection in hospitals (Gudlaugsson *et al.*, 2003; Pappas *et al.*, 2003). While a compromised immune function contributes to pathogenesis (Gow and Hube, 2012), it is less clear how *C. albicans* evolves to better exploit the host environment during the course of infection.

The total number of eukaryotic species on Earth has recently been estimated at 8.7 million, with fungi making up approximately 7% (611,000 species) of this number (Mora *et al.*, 2011). Of all fungi, only around 600 species are human pathogens. (Brown, Denning and Levitz, 2012) This relatively small group encompasses fungi that cause relatively mild infections of the skin (e.g., dermatophytes and *Malassezia* species), fungi that cause severe cutaneous infections (e.g., *Sporotrix schenkii*) and fungi that have the potential to cause life-threatening systemic infections (e.g., *Aspergillus fumigatus*, *Cryptococcus neoformans*, *Histoplasma capsulatum*, and *Candida albicans*). *C. albicans* can cause two major types of infections in humans: superficial infections, such as oral or vaginal Candidiasis, and life-threatening systemic infection.

C. albicans and to a lesser extent other *Candida* species are present in the oral cavity of up to 75% of the population (Ruhnke, 2002). In healthy individuals, this colonisation generally remains benign. However, mildly immunocompromised individuals can frequently suffer from recalcitrant infections of the oral cavity. These oral infections with *Candida* species are termed “oral Candidiasis” (OC). (Ruhnke, 2002) Such infections are predominantly caused by *C. albicans* and can affect the oropharynx and/or the oesophagus of persons with dysfunctions of the adaptive immune system. HIV is a major risk factor for developing OC. Further risk factors for developing OC include the wearing of dentures and extremes of age (Pappas *et al.*, 2009). It is estimated that approximately 75% of all women suffer at least once in their lifetime from vulvovaginal Candidiasis (VVC), with 40–50% experiencing at least one additional episode of infection (Hurley and De Louvois, 1979; Sobel, 2007). A small percentage of women (5–8%) suffer from at least four recurrent VVC per year. (Foxman *et al.*, 1998) Predisposing factors for VVC are less well defined than for OC and include diabetes mellitus, use of antibiotics, oral contraception, pregnancy and hormone therapy (Fidel, 2004). Despite their frequency and associated morbidity, superficial *C. albicans* infections are non-lethal. In stark contrast, systemic candidiasis is associated with a high crude mortality rate, even with first line antifungal therapy (Perlroth *et al.*, 2007; Pfaller and Diekema, 2007; Pfaller and Diekema, 2010). Both neutropenia and damage of the gastrointestinal mucosa are risk factors for the development of experimental systemic (disseminated) candidiasis. (Koh *et al.*, 2008) Further risk factors include central venous catheters, which allow direct access of the fungus to the bloodstream, the application of broad-spectrum antibiotic, which enable fungal overgrowth, and trauma or gastrointestinal surgery, which disrupts mucosal barriers (Spellberg *et al.*, 2012).

2.2 Clinical Manifestations of *Candida albicans*

i Cutaneous Candidiasis

Cutaneous candidiasis is a skin infection caused by *C. albicans* (Pfaller and Diekema, 2010). In immune-deficient patients such as those with innate anaemia and impaired natural immunity by physiological abnormalities in phagocytic cells, *Candida albicans* is a primary threat (Rezael *et al.*, 2007; Segal *et al.*, 2008). Cutaneous candidiasis is a well-defined disease characterised by *Candida albicans* infections of the skin (Espinosa-Hernández *et al.*, 2020). A considerable ratio of healthy individuals carries a detectable number colonising *Candida albicans* on the skin, oral cavity and gastrointestinal tract (Cannon *et al.*, 1995). They colonise mostly on the surface of oral cavity and are most frequently present on the dorsum of the tongue. Carriages of *Candida* species rarely establish into mucocutaneous candidiasis, so cutaneous and mucosal candidiasis poorly progresses to invasive candidiasis. *Candida* skin infections may be expressed as interdigital candidiasis between fingers and toes. *Candida albicans* develops in hot, dark and humid areas which explains why it occurs between fingers and toes. Fissuring, maceration, pimples and cysts are mostly present on interdiginious skin area. *Candida albicans* also causes rashes affecting anus and buttock areas usually known as diaper dermatitis. The rash can spread to the scrotum, inner thighs and gluteal area. *Candida albicans* is major source of folliculitis in immunosuppressed and obese patients (Maródi, 2014).

ii Oral Candidiasis

Oral candidiasis is an infection of the oral cavity by *Candida albicans*, first described in 1838 by paediatrician Francois Veilleux. The condition is generally obtained secondary to immune suppression, which can be local or systemic, including extremes of age (new-borns and elderly), immunocompromising diseases such as HIV/AIDS, and chronic systemic steroid and antibiotic use (Fangtham *et al.*, 2014; Millsop and Fazel, 2016). An example of local immunosuppression is inhaled corticosteroids, often prescribed in the preventive treatment of asthma and chronic obstructive pulmonary disease. Even though acute pseudomembranous candidiasis, also known as thrush, is the most common form of oral candidiasis, it is important to recognize that other types of oral candidiasis exist and that candidal infections can present as white and erythematous lesions. White lesions include acute pseudomembranous candidiasis and chronic hyperplastic vandidiasis whilst red lesions include acute and chronic erythematous candidiasis, angular cheilitis, median rhomboid glossitis and linear gingival erythema (Millsop and Fazel, 2016). Other rare oral types not included in these categories are cheilo-candidiasis, chronic mucocutaneous candidiasis, and chronic multifocal candidiasis (Millsop and Fazel, 2016). Diagnosis of oral candidiasis is often clinical, based on clinical examination, medical history taking, and assessment of risk factors. A biopsy is recommended for certain types in addition to empirical treatment. Cultures are usually done if antifungal treatment is ineffective. Topical antifungal therapy and oral hygiene measures are generally sufficient to resolve mild oral candidiasis and systemic antifungal therapy is usually reserved for patients who are refractory or intolerant to topical treatment and those at increased risk of developing systemic infections (Akpan and Morgan, 2002).

Oral candidiasis can occur in immunocompetent or immunocompromised patients but is more common in immunocompromised hosts. More than 90% of patients with HIV develop oral candidiasis at some point during the duration of the disease (Mahajan *et al.*, 2015). Oral candidiasis occurs equally in males and females. It typically occurs in neonates and infants, although it is rare in the first week of life. It is most common during the fourth week of life and less common in infants older than six months, likely secondary to the development of host immunity. Signs and symptoms of immunosuppression in these patients are diarrhoea, rashes, repeated infections and hepatosplenomegaly.

Candidal species cause oral candidiasis when a patient's host immunity becomes disrupted. This disruption can be local, secondary to oral corticosteroid use. Overgrowth of the fungus then leads to the formation of a pseudomembrane. Vaginal infections can colonise neonates as they pass through the birth canal. Alternatively, neonates and infants may contract the disease through colonised breasts when breastfeeding. A patient's oral *Candida* infection can often lead to gastrointestinal involvement and subsequent candidal diaper dermatitis. Candidal species thrive in moist environments. As such, females can develop vaginal candidiasis as well. In healthy patients, the patient's immune system and normal bacteria flora inhibit *Candida* growth. Consequently, immunosuppression such as diabetes, dentures, steroid use, malnutrition, vitamin deficiencies and recent antibiotic use often leads to the disease.

iii Invasive Candidiasis (IC)

Invasive candidiasis refers to bloodstream infections caused by *Candida* spp. Most often occur after passing through the intestinal barrier. For example, after surgery, it could spread to the abdominal cavity, and enter the bloodstream and cause candidiasis (Pappas *et al.*, 2018). *C. albicans* overgrowth can trigger an impairment of the immune response, leading to opportunistic infections in various organs, i.e., invasive candidiasis (Jabra-Rizk *et al.*, 2016). Therefore, the invasive disease is usually the result of increased or abnormal colonisation, along with local or generalised lack of host defence (Rosati *et al.*, 2020). Invasive candidiasis is not a single clinical entity but it is a disorder with countless clinical manifestations that can potentially affect any organ (Pappas *et al.*, 2018).

At least 15 different *Candida* spp. can cause infections in humans. However, five pathogens cause the most invasive infections: *Candida albicans*, *Candida glabrata, Candida tropicalis, Candida parapsilosis*, and *Candida krusei*, but the most common pathogen in the clinical setting is *C. albicans* (Pappas *et al.*, 2018).

Invasive *Candida* infections are most often associated with candidemia (*Candida* species in the blood), primarily in immunosuppressed patients and those requiring intensive care.

Immunosuppressed patients are at special risk for candidemia, including those with haematological malignancies (in patients who have just recovered from an episode of neutropenia), recipients of haematopoietic cell transplants or solid organ, and those given chemotherapeutic agents for a variety of different diseases. Other risk factors are associated with extensive gastrointestinal mucosal damage, broad-spectrum antibiotics and central venous catheters. *Candida albicans* is the most typical cause of candidiasis, although many *Candida* species other than *albicans* have been isolated in recent years (Yapar, 2014; Pappas *et al.*, 2015). When *C. albicans* enters the bloodstream, it can cause infections of the bladder and kidneys, endophthalmitis, meningitis, osteoarticular infections, endocarditis, peritonitis and intra-abdominal infections, pneumonia, empyema, mediastinitis and pericarditis.

Candiduria refers to the presence of *Candida* in the urine. Candiduria as a source of candidemia typically occurs in patients who have urinary tract abnormalities, most often urinary tract obstruction and/or in those who have undergone a urinary tract procedure. It is common in hospitalised patients, although it is often challenging to distinguish colonisation from a bladder infection. Renal transplantation was thought to be a risk factor for ascending infection and candidemia when candiduria was present. Studies show that more than 50% of hospitalised patients with candiduria have *Candida albicans* isolated (Ang *et al.*, 1993; Kuffman *et al.*, 2000).

C. albicans does not often cause clinically meaningful pneumonia in adults. Despite that, *Candida albicans* is often isolated from patients' respiratory tracts in intensive care units, intubated patients, or patients with a chronic tracheostomy. In most cases, this reflects colonisation of the airways and not an infection (Pappas *et al.*, 2015). *Candida* pneumonia has been noted in seriously immunocompromised patients with disseminated conditions, deficient birth weight new-borns and individuals with malignancies (Haron *et al.*, 1993; Kontoylannis *et al.*, 2002; Barton *et al.*, 2017). Since contamination issues confuse an antemortem diagnosis, a final diagnosis of invasive *Candida* pneumonia needs histological verification, which is typically achieved only at autopsy. A bronchoalveolar lavage is a diagnostic tool for verifying pneumonia and determining the causative pathogen (Schnabel *et al.*, 2014). *Candida* species infect bones and joints due to either haematogenous seeding or inoculation during trauma, intra-articular injection, a surgical procedure or injection drug use.

Osteoarticular infections often become symptomatic months or as long as a year after an episode of fungemia or a surgical procedure. The manifestations are generally subtler than bacterial infections at the same sites. Both of these factors contribute to long delays in diagnosis, especially in patients with vertebral osteomyelitis. The major symptoms of *Candida* arthritis are pain and decreased range of motion, whereas local pain is the predominant symptom of *Candida* osteomyelitis. Only one *Candida* colony is considered pathogenic in a biopsy or aspirate culture of joint fluid or bone (Johnson and Perfect, 2001; Miller and Mejicano, 2001; Gamaletsou *et al.*, 2012).

Candida infections of the central nervous system most typically affect the meninges (although they are all generally uncommon). This most often happens in premature infants. The infection may be secondary to haematogenous spread or direct inoculation. Predisposing factors include neurosurgery, newer antibiotics, and corticosteroids. Fever, meningismus, elevated cerebrospinal fluid pressure, and localising neurological signs are often present.

Candida albicans seems to be the most pathogenic *Candida* spp., leading to increased mortality rates in invasive infection when compared to other *Candida* species (Weitkamp and Nania, 2007; Shoham *et al.*, 2011). Fungal endocarditis represents 1–6% of the total spectrum of endocarditis. *Candida* endocarditis is one of the most severe candidiasis manifestations and is the most common cause of fungal endocarditis (Mamtani *et al.*, 2020). Due to the rarity of candidal infective endocarditis, the prognosis, epidemiology and optimal treatment of *Candida* infective endocarditis have been insufficiently described. Therapy procedures are obtained mainly from single-site case series and case reports. *Candida* endocarditis results from candidemia and is usually seen in patients with prosthetic heart valves, people who inject intravenous drugs, and in patients who have indwelling central venous catheters and prolonged fungemia (Baddley *et al.*, 2008).

Candida albicans (and other yeasts) can cause nosocomial infections, which involve the transmission by the hands of healthcare professionals or contaminated material (e.g., rinsing the central venous catheter with saline used for multiple patients) (Author *et al.*, 2015; Pappas *et al.*, 2018). Critical challenges in treating candidemia and invasive candidiasis include prevention, early detection, and rapid initiation of appropriate systemic antifungal therapy. The untimely initiation of treatment could result in poorer clinical outcomes (Calandra *et al.*, 2016; Logan *et al.*, 2020).

The emergence of antifungal resistance is a new problem globally, which is an additional concern in treating infections caused by *C. albicans* and other *Candida* sp (Wiederhold, 2017; Beardsley *et al.*, 2018; Costa-de-Oliveria, 2020). Published data suggest that fluconazole antifungal prophylaxis reduces the incidence of invasive candidiasis among high-risk ICU patients (Siddharthan *et al.*, 2016). However, it should be borne in mind that antifungal prophylaxis may favour resistance development (Wiederhold, 2017). Targeted antifungal prophylaxis is warranted in high-risk recipients of the liver, pancreas, small intestine or hematopoietic stem cells (Eschenauer *et al.*, 2015). The control of the infection source and early initiation of treatment with effective systemic antifungal therapy, usually before the diagnosis of invasive candidiasis is confirmed, is crucial for successfully treating invasive candidiasis (Hsu *et al.*, 2018; Pappas *et al.*, 2018).

iv Vulvovaginal Candidiasis (VVC)

Vulvovaginal candidiasis (VVC) is a common fungal infection caused by *Candida* species, predominantly *C. albicans* (Sobel, 1985). Historical reports approximate that 70% of all women will have at least one episode of VVC during their reproductive years (Sobel *et al.*, 1998). The pathological hallmark of the disease is an acute inflammatory condition of the vulva and vaginal mucosa induced by and accompanied with overgrowth of *Candida* organisms that normally exists as a quiescent vaginal commensal (Fidel *et al.*, 2004). Signs and symptoms of VVC are typically characterised by a white clumpy discharge, burning redness and itching in the vulva and vagina and dyspareunia (Sobel, 1992). The onset of most VVC cases is believed to be associated with a wide range of predisposing factors or triggering events including the use of antibiotics, increased estrogen levels (e.g. high estrogen oral contraceptives, hormone replacement therapies, pregnancy), uncontrolled diabetes mellitus, sexual activities and tight-fit clothing (Sobel, 1985; Sobel, 1997). In addition, an estimated 8−10% of women are susceptible to recurrent VVC (RVVC), having 4 or more episodes per annum (Sobel, 1992). Multiple recurring infections are often idiopathic without regard to the array of potential risk factors. Unlike most episodic or sporadic VVC, RVVC cases require maintenance regimens with long-term use of antifungals over several months or longer to avoid recurrence (Sobel, 1992; Achkar and Fries, 2010).

v Gastroentero-Candidiasis

The gastrointestinal flora is unique in each individual and mostly constant in healthy people. There are differences in its composition along the digestive tract, and it varies from person to person depending on dietary or hygienic habits and age (Faith *et al.*, 2013; Sekirov *et al.*, 2010; Basmaciyan *et al.*, 2019; Jeziorek *et al.*, 2019).

The entry of *C. albicans* into the bloodstream and the development of disseminated candidiasis mostly occur after an invasive infection of the gastrointestinal system (Zhu and Filler, 2010; Tong and Tang, 2017; Basmaciyan *et al.*, 2019). The main risk factors that promote the transition of *C. albicans* from a commensal or symbiotic to a pathogenic organism are dysbiosis of the residential microbiota, immune dysfunction, and damage to the muco-intestinal barrier (Allert *et al.*, 2018). At least one of four events is necessary for disseminated candidosis: direct invasion of epithelial cells (ECs) into blood capillaries and vessels, indirect translocation of *C. albicans* cells phagocytosed by host immune cells, direct damage of mucosal barriers and spread from fungal biofilms (Noble and Johnson, 2015; Allert *et al.*, 2018).

The most common type of infectious esophagitis caused by fungi is esophageal candidosis caused by *C. albicans* (Robertson *et al.*, 2013; Mohamed *et al.*, 2019). The risk of infection exists in immunosuppressed persons (HIV/AIDS-esophagitis caused by *C. albicans* is an AIDS-defining disease, haematological malignancies and transplant patients) and in patients with comorbidities such as diabetes, alcohol consumption and smoking, antibiotic therapy, glucocorticoids, chemotherapy, radiotherapy, upper esophageal damage and old age (Robertson *et al.*, 2013; Rosolowski *et al.*, 2013; Alsomali *et al.*, 2017; Hoversten *et al.*, 2019; Mohamed *et al.*, 2019). The clinical disease may be asymptomatic but is most commonly manifested by acute odynophagia, dysphagia and pain behind the sternum (Rosolowski *et al.*, 2013; Mushi *et al.*, 2018). Endoscopes show plaques of white to light yellow colour on the mucosa, which cannot be washed off and after their removal, the mucosa is red and ulcerative (Rosolowski *et al.*, 2013; Mohamed *et al.*, 2019). Esophageal candidosis, unlike oropharyngeal, should always be treated with systemic rather than topical antifungals. Three groups of drugs can be used in therapy: nystatin, amphotericin B and azole antifungals (most commonly fluconazole), where the choice depends on the degree of immunosuppression (Hoversten *et al.*, 2018; Mohamed *et al.*, 2019). Esophagitis caused by *C. albicans* is most often superficial, but complications and invasion with haematogenous dissemination (fungemia) are also possible and can subsequently lead to infection of other organs (Mohamed *et al.*, 2019).

Fungal diseases of the gastro-duodenum are less commonly reported. Mostly they occur as a secondary infection of individuals with tumours in this area, and they infiltrate benign or malignant ulcers that have a reduced ability to heal. Endoscopically, this looks like a white or grayish deposit that separates easily from the mucosa and is located at the base of the ulcer.

The ulcer mostly heals with antiulcer therapy (Castelo et sl., 2017). Possible intestinal infection may be superficial when the invasion is limited to the mucosa and submucosa, but can also be deep, where penetration is unlimited, and tissue destruction and perforation of the intestinal wall or spread to distant places occurs. Fungal infections are most commonly associated with inflammatory bowel disease (IBD). Predisposing factors are mucosal damage, mostly caused by surgery and chemotherapy, and impaired neutrophil function due to tumour therapy or long-term glucocorticoid use. When administering TNFα, *C. albicans* should be suspected if infections are detected early during IBD treatment (Stamatiades *et al.*, 2018).

The interaction of *C. albicans* as a pathogen with the intestinal mucosa occurs in the form of adhesion, invasion, damage and apoptosis. The main role in infection and consequently pathogenicity, is played by substances secreted by the fungal hyphae (Moyes *et al.*, 2015; Tong and Tand, 2017; Allert *et al.*, 2018). Increased colonisation and infection increase the secretion of antimicrobial peptides (AMPs) by host cells, but *C. albicans* has developed mechanisms to avoid their activity as the first step in adherence to intestinal epithelial cells (IEC). In addition to defending against AMPs, *C. albicans* must break down the mucus's protective layer to reach the epithelial cell layer. After adhering to the mucins, it secretes mucinolytic enzymes. After the first contact with the IEC, most fungal cells convert to the hyphal form and express genes that promote adhesion by releasing adhesins and hyphal invasions. The release of surface molecules, i.e., adhesins, is crucial in the process of adhesion to the host tissue. It can also adhere to enterocytes via polysaccharide molecules on the cell wall surface (Naglik *et al.*, 2017; Richardson *et al.*, 2018; Gil-Bona *et al.*, 2018; Basmaciyan *et al.*, 2019). Invasion by *C. albicans* takes place through two mechanisms, namely endocytosis and active penetration. Endocytosis is a host-driven process that does not require sustainable hyphae and occurs in the first four hours of interaction. Active penetration into the IEC is a procedure that requires sustainable forms of the fungus but does not require host activity and depends on the type of epithelial cells. It is thought that penetration takes place by the combination of mechanical pressure created by progressive elongation of the hyphae and lytic activity. This procedure allows *C. albicans* entry into enterocytes while the intercellular junctions remain intact. Unlike oral cells, fungi need specific genes to invade enterocytes (Allert *et al.*, 2018; Basmaciyan *et al.*, 2019). Invasion od enterocytes by *C. albicans* contributes to IEC damage and cell death. It is assumed that necrotic cell death is the main mechanism of damage observed during translocation of *C. albicans* through enterocytes. After the hyphae form, a cytolytic toxin called candidalysin is secreted, which is responsible for damaging host cells. It is the first peptide toxin identified in human fungal pathogens (Basmaciyan *et al.*, 2019). Antifungal therapy should be utilised in patients with intra-abdominal infections and risk factors for candidiasis, such as recent abdominal surgery, necrotizing pancreatitis or anastomotic leakage.

2.3 Pathogenicity of *Candida albicans*

The ability of *C. albicans* to infect such diverse host niches is supported by a wide range of virulence factors and fitness attributes. A number of attributes, including the morphological transition between yeast and hyphal forms, the expression of adhesins and invasins on the cell surface, thigmotropism, the formation of biofilms, phenotypic switching and the secretion of hydrolytic enzymes are considered virulence factors. Additionally, fitness attributes include rapid adaptation to fluctuations in environmental pH, metabolic flexibility, powerful nutrient acquisition systems and robust stress response machineries (Perez and Johnson, 2013).S

Pathogenicity Mechanisms

i Polymorphism

C. albicans is a polymorphic fungus that can grow either as an ovoid-shaped budding yeast, as elongated ellipsoid cells with constrictions at the septa (pseudohyphae) or as parallel-walled true hyphae. (Berman and Sudbery, 2002) Further morphologies include white and opaque cells, formed during switching and chlamydospores, which are thick-walled, spore-like structures. (Sudbery, Gow and Berman, 2003) While yeast and true hyphae are regularly observed during infection and have distinct functions, the role of pseudohyphae and switching in vivo is rather unclear and chlamydospores have not been observed in patient samples (Staib and Morschhäuser, 2007; Soll, 2009). A range of environmental cues affect *C. albicans* morphology. For example, at low pH (< 6) *C. albicans* cells predominantly grow in the yeast form, while at a high pH (> 7) hyphal growth is induced (Odds, 2011). A number of conditions, including starvation, the presence of serum or N-acetylglucosamine, physiological temperature and carbon dioxide promote the formation of hyphae (Sudbery, 2011). Morphogenesis has also been shown to be regulated by quorum sensing, a mechanism of microbial communication (Albuquerque and Casadevall, 2011). In *C. albicans*, the main quorum sensing molecules include farnesol, tyrosol and dodecanol (Hornby *et al.*, 2001; Hall *et al.*, 2011).

ii Adhesins and Invasions

C. albicans has a specialised set of proteins (adhesins) which mediate adherence to other *C. albicans* cells, to other microorganisms, to abiotic surfaces and to host cells (Garcia *et al.*, 2011; Verstrepen and Klis, 2006). Arguably the best studied *C. albicans* adhesins are the agglutinin-like sequence (ALS) proteins which form a family consisting of eight members

(Als1–7 and Als9). The ALS genes encode glycosylphosphatidylinositol (GPI)-linked cell surface glycoproteins. Of the eight Als proteins, the hypha-associated adhesin Als3 is especially important for adhesion (Zordan and Cormack, 2012; Phan *et al.*, 2007; Murciano *et al.*, 2012). ALS3 gene expression is upregulated during infection of oral epithelial cells in vitro and during in vivo vaginal infection (Wächtler *et al.*, 2011; Zkikhany *et al.*, 2007; Naglik *et al.*, 2011; Cheng *et al.*, 2005). Another important adhesin of *C. albicans* is Hwp1, which is a hypha-associated GPI-linked protein. Hwp1 serves as a substrate for mammalian transglutaminases and this reaction may covalently link *C. albicans* hyphae to host cells (Zordan and Cormack, 2012; Staab *et al.*, 1999; Sundstorm and Balish, 2002).

iii Biofilm Formation

A further important virulence factor of *C. albicans* is its capacity to form biofilms on abiotic or biotic surfaces. Catheters, dentures (abiotic) and mucosal cell surfaces (biotic) are the most common substrates. Biofilms form in a sequential process including adherence of yeast cells to the substrate, proliferation of these yeast cells, formation of hyphal cells in the upper part of the biofilm, accumulation of extracellular matrix material and, finally, dispersion of yeast cells from the biofilm complex (Finkel and Mitchell, 2011). Mature biofilms are much more resistant to antimicrobial agents and host immune factors in comparison to planktonic cells (Fanning and Mitchell, 2011; Finkel and Mitchell, 2011). The factors responsible for heightened resistance include the complex architecture of biofilms, the biofilm matrix, increased expression of drug efflux pumps and metabolic plasticity (Fanning and Mitchell, 2011). Dispersion of yeast cells from the mature biofilm has been shown to directly contribute to virulence, as dispersed cells were more virulent in a mouse model of disseminated infection. (Uppuluri *et al.*, 2010) The major heat shock protein Hsp90 was recently identified as a key regulator of dispersion in *C. albicans* biofilms (Robbins *et al.*, 2011). In addition, Hsp90 was also required for biofilm antifungal drug resistance (Robbins *et al.*, 2011). Several transcription factors control biofilm formation. These include the transcription factors Bcr1, Tec1 and Efg1 (Fanning and Mitchell, 2011).

In a recent study, Nobile and co-workers (2009) investigated the transcriptional network regulating biofilm formation and identified further, previously unknown regulators of biofilm production (Nobile *et al.*, 2011). These novel factors include Ndt80, Rob1 and Brg1. Deletion of any of these regulators (BCR1, TEC1, EFG1, NDT80, ROB1 or BRG1) resulted in defective biofilm formation in in vivo rat infection models (Nobile *et al.*, 2011). Extracellular matrix production is controlled by additional factors. The zinc-responsive transcription factor Zap1 negatively regulates β-1,3-glucan, the major component of biofilm matrix (Nobile *et al.*, 2009). Glucoamylases (Gca1 and Gca2), glucan transferases (Bgl2 and Phr1) and the exo-glucanase, Xog1, are positive regulators of β-1,3 glucan production (Nobile *et al.*, 2009; Taff *et al.*, 2009). While expression of GCA1 and GCA2 are controlled by Zap1, the enzymes Bgl2, Phr1 and Xog1 function independently of this key negative regulator. (Taff *et al.*, 2009) Biofilms formed by mutants lacking BGL2, PHR1 or XOG1 were shown to be more susceptible to the antifungal agent, fluconazole, both *in vitro* and *in vivo* (Taff *et al.*, 2009). Furthermore, recent studies indicate that *C. albicans* biofilms are resistant to killing by neutrophils and do not trigger production of reactive oxygen species (ROS). (Xie *et al.*, 2012). Evidence suggests that β-glucans in the extracellular matrix protect *C. albicans* from these attacks (Xie *et al.*, 2012).

iv Contact Sensing and Thigmotropism

An important environmental cue that triggers hypha and biofilm formation in *C. albicans* is contact sensing. Upon contact with a surface, yeast cells switch to hyphal growth (Kumamoto, 2008). On certain substrates, such as agar or mucosal surfaces, these hyphae can then invade into the substratum. Contact to solid surfaces also induces the formation of biofilms (Kumamoto, 2008). On surfaces with particular topologies (such as the presence of ridges) directional hyphal growth (thigmotropism) may occur (Brand *et al.*, 2006) Brand *et al.* (2006) demonstrated that thigmotropism of *C. albicans* hyphae is regulated by extracellular calcium uptake through the calcium channels Cch1, Mid1 and Fig1. Additional mechanisms include the polarisome Rsr1/Bud1-GTPase module (Brand and Gow, 2009). Brand *et al.* (2008) also provided evidence that *C. albicans* thigmotropism is required for full damage of epithelial cells and normal virulence in mice. Therefore, the correct sensing and response to both abiotic (biofilm formation) and biotic (invasion) surfaces is important for pathogenicity. (Brand *et al*., 2008)

v Secreted Hydrolases

Following adhesion to host cell surfaces and hyphal growth, *C. albicans* hyphae can secrete hydrolases, which have been proposed to facilitate active penetration into these cells (Wächtler *et al.*, 2012). In addition, secreted hydrolases are thought to enhance the efficiency of extracellular nutrient acquisition (Naglik *et al.*, 2003). Three different classes of secreted hydrolases are expressed by *C. albicans*: proteases, phospholipases and lipases. The family of secreted aspartic proteases (Saps) comprises ten members, Sap1–10. Sap1–8 are secreted and released to the surrounding medium, whereas Sap9 and Sap10 remain bound to the cell surface (Naglik *et al.*, 2003; Taylor *et al.*, 2005; Albrecht *et al.*, 2006). Sap1–3 have been shown to be required for damage of reconstituted human epithelium (RHE) in vitro, and for virulence in a mouse model of systemic infection (Schaller *et al.*, 1999; Hube *et al.*, 1997). However, the relative contribution of Saps to *C. albicans* pathogenicity is controversial. Recent results indicate that Saps are not required for invasion into RHE and that Sap1–6 are dispensable for virulence in a mouse model of disseminated candidiasis (Lermann and Morschhäuser, 2008; Correia *et al.*, 2010). However, the observed expansion of Sap-encoding genes in *C. albicans* compared with its less pathogenic relatives suggests a role for these proteases in virulence (Moran *et al.*, 2012). The large size of the Sap family itself makes it likely that a certain degree of functional redundancy may exist. The family of phospholipases consists of four different classes (A, B, C and D). (Niewerth and Korting, 2001). Only the five members of class B (PLB1–5) are extracellular and may contribute to pathogenicity via disruption of host membranes (Mavor *et al.*, 2005). The third family of secreted hydrolases, the lipases, consists of 10 members (LIP1–10) (Fu *et al.*, 2000; Hube *et al.*, 2000).

vi pH-Sensing and Regulation

In the human host, *C. albicans* is exposed to a surrounding pH ranging from slightly alkaline to acidic (Davis, 2009). Additionally, depending on the host niche, the environmental pH can be very dynamic. Therefore, *C. albicans* must be able to adapt to changes in pH (Davis, 2009). The pH of human blood and tissues is slightly alkaline (pH 7.4), while the pH of the digestive tract ranges from very acidic (pH 2) to more alkaline (pH 8), and the pH of the vagina is around pH 4 (Davis, 2009). Neutral to alkaline pH can cause severe stress to *C. albicans*, including malfunctioning of pH-sensitive proteins and impaired nutrient acquisition (as a consequence of a disrupted proton gradient) (Davis, 2009). Among the first proteins identified as being important for adaptation to changing pH were the two cell wall β-glycosidases Phr1 and Phr2. (Fonzi, 1999) PHR1 is expressed at neutral-alkaline pH. In contrast, PHR2 is mainly expressed at acidic pH (Mühlschlegel and Fonzi, 1997). Correspondingly, Phr1 is required for systemic infections and Phr2 is essential for infections of the vagina (De Bernadis *et al.*, 1998). *C. albicans* senses pH via the Rim101 signal transduction pathway. In this pathway, environmental pH is gauged by the plasma membrane receptors Dfg16 and Rim21 (Davis, 2009). Activation of these receptors leads to induction of a signalling cascade, finally leading to activation of the major pH-responsive transcription factor Rim101 via its proteolytic cleavage. Rim101 then enters the nucleus and mediates pH-dependent responses (Davis, 2009).

vii Metabolic Adaptation

Nutrition is a central and fundamental prerequisite for survival and growth of all living organisms. Metabolic adaptability mediates the effective assimilation of alternative nutrients in dynamic environments (Brown *et al.*, 2012). This metabolic flexibility is particularly important for pathogenic fungi during infection of different host niches (Fleck *et al.*, 2011; Brock, 2012). Glycolysis, gluconeogenesis and starvation responses are all thought to contribute to host colonisation and pathogenesis, but their specific contribution may be highly niche-specific and is still only partially understood. In healthy individuals, *C. albicans* is predominantly found as part of the gastrointestinal microbiome. Although the concentration of nutrients in this environment can be naturally high, growth of the fungus is believed to be controlled through competition with other members of the intestinal microbial flora (Brock, 2012). During disseminated candidiasis in susceptible individuals, *C. albicans* gains access to the bloodstream. Blood is relatively rich in glucose (6–8 mM), (Brock, 2012) the preferred nutrient source of most fungi. However, phagocytic cells (macrophages and neutrophils) can efficiently phagocytose *C. albicans*. Once inside a macrophage or neutrophil, however, the nutritional environment completely changes for the fungus. Not only does the phagocyte produce highly reactive intermediates like ROS, reactive nitrogen species (RNS) and antimicrobial peptides (AMPs), it also restricts the availability of nutrients, thereby creating an environment of nutrient starvation (Frohner *et al.*, 2009). Prompt and efficient metabolic plasticity is therefore required for adaptation of *C. albicans* to such a hostile host milieu. Inside macrophages, the fungus initially switches from glycolysis to gluconeogenesis and a starvation response (activation of the glyoxylate cycle). Lipids and amino acids are proposed to serve as nutrient sources within macrophages (Lorenz *et al.*, 2004). In addition to metabolic flexibility, the fungus has also evolved ways to escape from macrophages by inhibiting the production of antimicrobial effectors and inducing hyphal formation. Hyphae formed inside phagocytic cells can pierce through the host immune cell by mechanical forces and can permit escape (Lorenz *et al.*, 2004; Gosh *et al.*, 2009). During systemic candidiasis, fungal cells can disseminate to virtually every organ within the human host, each with potentially different availability of nutrients. In the liver for example, *C. albicans* has access to large quantities of glycogen, the main storage molecule of glucose. The brain has high concentrations of glucose and vitamins as potential nutrient sources (Fleck *et al.*, 2011). In other tissues, *C. albicans* faces relatively poor glucose concentrations and uses alternative metabolic pathways to utilise host proteins, amino acids, lipids and phospholipids. The fungus can use secreted proteases to hydrolyse host proteins. It was recently shown that adaptation to different nutrient sources by *C. albicans* not only promotes survival and growth, but also affects virulence (Ene *et al.*, 2012). Growth on alternative carbon sources, such as lactate or amino acids, rendered the fungus more resistant to environmental stresses and increased its virulence potential in both a mouse model of systemic candidiasis and a murine vaginal infection model (Wysong *et al.*, 1998) Furthermore; the glyoxylate cycle has been shown to be required for full virulence in *C. albicans*. (Lorenz and Fink, 2001) Uptake of amino acids, and likely also polyamines, affects the virulence of *C. albicans* by allowing the fungus to auto-induce hypha formation through extracellular alkalinisation (Vylkova *et al.*, 2011; Mayer *et al.*, 2012). In summary, during infection, the main nutrient sources for *C. albicans* are likely to be host-derived glucose, lipids, proteins and amino acids, depending on the anatomical niche. Besides being able to use these different nutrients individually, the ability of *C. albicans* to rapidly and dynamically respond to host and pathogen-induced changes in micro-environmental nutrient availability contributes to its success as a pathogen.

viii Environmental Stress Response

A robust stress response contributes to the survival and virulence of *C. albicans* by facilitating the adaptation of the fungus to changing conditions and protecting it against host-derived stresses. Phagocytic cells of the immune system produce oxidative and nitrosative stresses. pH-stress occurs, for example, in the gastrointestinal and urogenital tract (Brown *et al.*, 2012). Stress-responsive regulatory pathways, as well as downstream targets, were shown to be essential not only for efficient stress adaptation, but also for full virulence of the fungus (Brown *et al.*, 2012). In fact, several mutants lacking genes encoding regulators of stress response or detoxifying enzymes are attenuated in virulence. Cellular responses to stresses include heat shock-, osmotic-, oxidative- and nitrosative-stress responses (Brown *et al.*, 2012). The heat shock response is mediated by heat shock proteins which act as molecular chaperones to prevent deleterious protein unfolding and aggregation. Additionally, thermal stress leads to trehalose accumulation in *C. albicans*, which is thought to act as a “chemical chaperone” by stabilising proteins prone to unfolding (Brown *et al.*, 2012). However, the exact function of trehalose accumulation following thermal insults remains unknown. The osmotic stress response results in intracellular accumulation of the compatible solute glycerol to counteract loss of water due to the outward-directed chemical gradient. Glycerol biosynthesis is mediated by the glycerol 3-phosphatase (Gpp1) and the glycerol 3-phosphate dehydrogenase (Gpd2) (Wächtler *et al.*, 2011).

ix Heat Shock Proteins

The heat shock response is a conserved reaction of living organisms to stressful conditions such as high temperature, starvation and oxidative stress (Lindquist, 1986; Lindquist, 1992). Such stresses can induce protein unfolding and nonspecific protein aggregation, ultimately leading to cell death. In order to prevent this, cells produce heat shock proteins (Hsps) (Lindquist, 1992). These specialised proteins act as chaperones and prevent protein unfolding and aggregation by binding to their clients and stabilising them (Richter *et al.*, 2010). Six major Hsps have been identified in *C. albicans*: Hsp104, Hsp90, Hsp78, two Hsp70 proteins (Ssa1 and Ssa2) and Hsp60. *HSP104* encodes a Hsp required for proper biofilm formation and virulence in a *Caenorhabditis elegans* infection model. (Fiori *et al.*, 2012) Hsp90 is a major Hsp in *C. albicans* and regulates drug resistance, morphogenesis, biofilm formation and virulence (Cowen *et al.*, 2009; Shapiro *et al.*, 2009; Singh *et al.*, 2009; LaFayette *et al.*, 2010; Robbins *et al.*, 2011). *HSP78* encodes an uncharacterised Hsp that is transcriptionally upregulated in response to phagocytosis by macrophages (Lorenz *et al.*, 2004). The two *C. albicans* Hsp70 family members, Ssa1 and Ssa2 (stress-70 subfamily A), are expressed on the cell surface and function as receptors for antimicrobial peptides, e.g., Ssa2 binds histatin 5 (Li *et al.*, 2003; Li and Sun, 2006; Sun *et al.*, 2008). Ssa1 also acts as an invasin (Sun *et al.*, 2010). (Sun *et al.*, 2010) Finally, *HSP60* encodes a putative mitochondrial Hsp of unknown function.

x Small Heat Shock Proteins

In addition to the above mentioned heat shock proteins, six small Hsps (sHsps) have also been identified in *C. albicans* (Inglis *et al.*, 2012). Hsps are low-molecular-mass chaperones that prevent protein aggregation (Naberhaus, 2002; Haslbeck *et al.*, 2002). Upon heat, or other forms of stress, cells express sHSPs which transition from an oligomeric to a multimeric state and bind aggregated proteins (Eyles *et al.*, 2010). In these chaperone-aggregate complexes, client proteins are held ready for disaggregation and refolding by other major Hsps, such as Hsp104 (Cashikar *et al.*, 2005). *C. albicans* is predicted to encode six sHsps: Hsp31, Hsp30, Hsp21, two Hsp12 proteins and Hsp10. (Inglis *et al.*, 2012) As yet, only Hsp12 and Hsp21 have been investigated. Hsp12 is expressed in response to different stresses, including heat shock and oxidative stress. Deletion of both *HSP12* genes did not influence virulence of *C. albicans* in a *Drosophila* infection model (Fu *et al.*, 2012).

xi Metal Acquisition

Trace metals are essential for the growth and survival of all living organisms including humans, animals, plants, bacteria and fungi. Among the most important metals are iron, zinc, manganese and copper, all of which are essential for the proper function of a large number of proteins and enzymes. Pathogenic microorganisms, as well as their respective hosts, have evolved elaborate mechanisms to acquire or restrict access to these metals (Hood and Skaar, 2012). To date, the most widely investigated transition metal with regard to pathogenesis is iron. *C. albicans* acquires this metal by different strategies, including a reductive system, a siderophore uptake system and a heme-iron uptake system (Almeida *et al.*, 2009). The reductive system mediates iron acquisition from host ferritin, transferrin or the environment. The adhesin and invasin Als3 was shown to be the receptor for ferritin (Almeida *et al.*, 2008). The heme-iron uptake system promotes iron acquisition from haemoglobin and hemeproteins and is mediated by the heme-receptor gene family members *RBT5, RBT51, CSA1, CSA2 and PGA7* (RBT6) (Weissman and Kornitzer, 2004; Almeida *et al.*, 2009). The role of the other four heme-binding proteins (Rbt51, Csa1, Csa2 and Pga7) in virulence has not yet been investigated. Zinc is the second most abundant metal in most living organisms. (Hood and Skaar, 2012) *C. albicans* secretes the zinc-binding protein Pra1 (pH-regulated antigen 1), which, analogous to siderophore-mediated iron acquisition, acts as a zincophore by binding extracellular zinc and re-associating with the fungal cell. Re-association of Pra1 is mediated by the zinc transporter Zrt1 (Citiulo *et al.*, 2012). A putative manganese transporter, Ccc1, (Inglis *et al.*, 2012) and a copper transporter, Ctr1, have been identified, although their roles in virulence have not yet been determined.

2.4 Antifungal Resistance

Candidiasis is usually diagnosed late. Firstly, fungal infections are generally considered only after antibiotic treatments fail to reduce fever. Secondly, standard diagnostic approaches require blood cultures, which are slow and can have a low sensitivity. For instance, some studies reported sensitivities as low as 17% (Nguyen *et al.*, 2012) or 45% (Fortún *et al.*, 2014). Furthermore, although the four most common *Candida* (*C. albicans*, *C. glabrata, C. parapsilosis*, and *C. tropicalis*) can account for more than 80% of the cases, there is a long list with over 30 *Candida* that have been identified as candidemia agents (Gabaldón *et al.*, 2016), and the list keeps expanding. Added to the difficulty of a fast and accurate diagnosis, doctors face severe limitations with regards to treatment options. Currently, there are only four major classes of antifungals in clinical use: azoles, polyenes, echinocandins, and pyrimidine analogs (Odds *et al.*, 2016). This situation alarmingly decreases the chances of a successful treatment and increases the possibilities of a fatal outcome if the infecting pathogen is resistant to one or multiple drugs. Limitations in diagnostic methods further enhance the problems of a few therapeutic options, as different species may show diverse resistance profiles. Thus, diagnostics of the infecting agent, along with susceptibility tests, should be used to inform the choice of therapy. Over the last years, the intensive use of some antifungal drugs, such as azoles, has promoted a shift in the epidemiology of candidiasis, in which the incidence of *C. albicans* has decreased in favour of other species that are naturally less susceptible to this drug, such as *C. glabrata*. To the problem of the intrinsic variation of drug susceptibility among different *Candida*, we need to add the emerging issue of acquired resistance, which refers to the ability of yeasts to evolutionarily develop mechanisms that lower their susceptibility towards a given drug (Fraimow and Abrutyn, 1995). This process generally involves mutations ranging from chromosomal re-arrangements to point mutations. These mutations can affect drug resistance in different ways, ranging from directly interfering with the binding of the drug to its target to inducing gene expression changes that promote physiological states that reduce drug susceptibility. In this regard, an enhanced capacity to form biofilms can result in the acquisition of resistance, as these structures promote yeast survival upon exposure to the drug (Sardi *et al.*, 2013; Rodrigues *et al.*, 2017). The emergence of resistant strains, including those becoming resistant to multiple drugs, has been increasingly reported in recent years (Pfaller *et al.*, 2011; Pfaller *et al.*, 2014; Locahart *et al.*, 2017). In addition, it has been demonstrated that such resistant phenotypes can develop over the course of an infection, and in response to treatment, which adds yet another threat to patients (Pfaller, 2012). Despite the clinical and economic relevance of drug resistance in the context of yeast infections, this subject remains poorly studied, at least in comparison with the similar issue of antibiotic resistance in bacterial pathogens. Although parallels can be established, the evolutionary mechanisms underlying the emergence of resistance in fungi and bacteria are markedly different. While drug resistance in bacteria generally involves the transference, between strains or species, of genetically mobile elements such as genomic islands (Mohammad, 2014), in fungi, resistance commonly appears via genetic alterations within a lineage. Still, we are far from having a broad understanding of how resistance towards antifungal drugs emerges in the context of infection or commensalism in yeast pathogens. Fortunately, recent developments in sequencing technologies are enabling us to catalog and trace the origins of mutations conferring resistance to antifungal drugs in different species.

2.4.1 Mechanisms of Antifungal Resistance

Mechanisms of antifungal resistance in *Candida albicans* include:

- i Alterations in drug targets
- ii Overexpression of efflux pumps
- iii Genome Modification

2.4.1.1 Alterations in Drug Targets

i. Azoles

Azoles are heterocyclic compounds containing at least one nitrogen atom as part of the ring. Common azoles used as antimycotic agents include the triazoles: fluconazole, voriconazole, and posaconazole. These drugs act by targeting the cytochrome P450 enzyme, lanosterol 14α-demethylase that converts lanosterol to ergosterol. In yeast, this enzyme is encoded by the ERG11 gene. Similar to cholesterol in animals, ergosterol is the main membrane sterol in most fungal species, holding an important role in controlling membrane fluidity (Weete *et al.*, 2010). As a result of the action of azoles, the *Candida* cell membrane is depleted of ergosterol and accumulates other toxic 14α-methylated sterols. Subsequently, this causes the decrease in membrane fluidity and, in most of the cases, inhibits cell growth (Sheehan *et al.*, 1999). Fluconazole is the azole drug most widely used for the treatment of *Candida* infections. Its utility is attributed to its high bioavailability, high water solubility, and low affinity to plasma proteins (Andriole, 2000). Unfortunately, the fungistatic character of fluconazole and its extended, perhaps excessive use is inevitably leading to an increasing selection in favour of resistant yeast isolates.

In *C. albicans*, a common mechanism of azole resistance involves amino acid substitutions in the drug target, Erg11, which leads to lower drug-binding affinity for the lanosterol demethylase enzyme (Lamb *et al.*, 1997; Revie *et al.*, 2018). Over 140 amino acid substitutions in Erg11 have been associated with azole resistance in this species (Morio *et al.*, 2010) with the majority of these substitutions clustered into hot-spot regions ranging from amino acids 105–165, 266–287 and 405–488 (Marichal *et al.*, 1999). Notably, only a few amino acid substitutions have been confirmed experimentally to confer azole resistance. (Xiang *et al.*, 2013; Wu *et al.*, 2017) Molecular mapping of *C. albicans* ERG11 variant positions revealed that mutations reside in the catalytic site of the enzyme, the fungus-specific external loop and the proximal surface, as well as between the proximal surface and the heme. (Flowers *et al.*, 2015)

Overexpression of ERG11 is common in azole-resistant clinical isolates of *C. albicans* and directly contributes to increased target abundance, ultimately lowering drug susceptibility (Revie *et al*, 2018; Robbins *et al.*, 2017). For *C. albicans*, the transcriptional activator, Upc2, is a crucial regulator of many ergosterol biosynthesis genes, including ERG11. Gain-of-function (GOF) mutations in UPC2 cause the constitutive overexpression of ergosterol biosynthesis genes, a higher ergosterol content and a reduction in fluconazole susceptibility (MacPherson *et al.*, 2005; Silver *et al.*, 2004; Dunkei *et al.*, 2008; Heolmann *et al.*, 2010; Flowers *et al.*, 2012). Furthermore, disruption of UPC2 in an azole-resistant clinical isolate abrogated ERG11 overexpression and increased fluconazole susceptibility, highlighting the importance of this transcriptional activator in mediating *C. albicans* azole resistance (Vasicek *et al.*, 2014).

ii. Polyenes

Polyenes are poly-unsaturated organic compounds that contain at least three alternating double and single carbon–carbon bonds. Their antimycotic action is mediated by direct binding to and removal of ergosterol present in the fungal cell membrane. This results in the loss of membrane permeability, subsequent membrane leakage and eventually cell death (Kathiravan *et al*, 2012). In the 1950s, the polyene amphotericin B deoxycholate was the first approved successful antifungal drug (Perfect, 2017). Nowadays, amphotericin B continues to be broadly used despite its high toxicity, which results from structural similarities between ergosterol and human cholesterol. Due to this toxicity, the use of amphotericin B in high concentrations may be harmful and cause damage to human tissues, such as the kidneys (Groll *et al.*, 1998).

Resistance to polyenes is extremely uncommon; however, in the rare incidence that it does occur, it is mediated by alterations in enzymes that reduce drug-binding affinity or deplete ergosterol from the membrane (Ellis, 2002). In *C. albicans*, reduced amphotericin B susceptibility can occur through mutations in several ergosterol biosynthesis enzymes, including ERG2, (Jensen *et al.*, 2015) ERG3, (Martel *et al.*, 2010) ERG5, (Martel *et al.*, 2010) and ERG11 (Sanglard *et al.*, 2003). Aside from genetic alterations, azole treatment has also been attributed to enhanced polyene resistance in *Candida* species, as a consequence of azole-mediated reduction in cellular ergosterol levels (Vazquez *et al.*, 1998).

iii. Echinocandins

Echinocandins are amphiphilic lipopeptides and products of cyclopentamine. They can be formed during the fermentation of some fungi such as *Zalerion arboricola or Aspergillus nidulans var. echinulatus*, but nowadays, they are produced semi-synthetically for clinical use. The most common representatives of this class of drugs are caspofungin, micafungin, and anidulafungin. Echinocandins inhibit the biosynthesis of an essential component of the fungal cell wall, the 1,3-ß-glucan. In *Candida*, they target two subunits of the 1,3-ß-glucan synthase, encoded by the FKS1 and FKS2 genes (Sucher *et al.*, 2009), and eventually cause cell lysis. For most *Candida* species, echinocandin resistance is primarily mediated by mutations in the FKS genes. In *C. albicans*, mutations that confer echinocandin resistance occur in the essential gene, FKS1. Hot-spot regions correspond to amino acids 641–649 (hot spot 1) and amino acids 1357–1364 (hot spot 2) (Perlin, 2007). Mutations at these regions decrease the IC50 of the glucan synthase enzyme by several orders of magnitude, elevate MIC values and result in cross-resistance to diverse echinocandins (Park *et al.*, 2005; Garcia-Effron *et al.*, 2008; Garcia-Effron *et al.*, 2009; Garcia-Effron *et al.*, 2009).

2.4.1.2 Overexpression of Efflux Pumps

i. Azole

The upregulation of plasma membrane efflux pumps is a major mechanism governing azole resistance in many fungal pathogens. The first main class of efflux pumps implicated in azole drug resistance is the ATP-binding cassette (ABC) superfamily. ABC transporters possess two transmembrane-spanning domains and two cytoplasmic nucleotide-binding domains (NBDs). The NBD drives the movement of substrates across the fungal membrane via ATP hydrolysis (Coleman and Mylonakis, 2009). The second class of efflux pumps implicated in azole resistance is the major facilitator (MF) superfamily. Like the ABC superfamily, MF transporters also possess transmembrane-spanning helices but use the proton gradient generated across the plasma membrane to drive MF-mediated translocation (Coleman and Mylonakis, 2009).

In *C. albicans*, overexpression of two homologous ABC transporters, Cdr1 and Cdr2, have been frequently implicated in azole resistance, particularly in patients receiving long-term antifungal therapy (Robbins *et al.*, 2017; Revie *et al.*, 2018). Of the two transporters, Cdr1 is the major determinant of azole resistance; deletion of CDR1 in a clinical isolate reduces resistance to the azoles by 4- to 8-fold. However, deletion of CDR2 typically results in a much weaker effect (Tsao *et al.*, 2009). The transcription factor Tac1 (transcriptional activator of CDR) binds to cis-acting drug-response elements (DREs) in the promoters of CDR1 and CDR2 to regulate their expression.

ii. Polyenes and Echinocandins

In contrast to azoles, the upregulation of efflux pumps plays a rather insignificant role in conferring resistance to the echinocandins. This is consistent with the fact that echinocandins function at the outer leaflet of the fungal cell membrane, (Robbins *et al.*, 2016; Perfect, 2017) as well as the fact that these large cyclic hexapeptide molecules are likely to be poor substrates of fungal efflux pumps due to their size and hydrophobicity (Cannon *et al.*, 2009). Similarly, for polyenes, the lack of efflux-mediated mechanisms of resistance may be attributed to their inability to enter the drug-binding pocket of transporters (Cannon *et al.*, 2009).

2.4.1.3 Genome Modification

Fungal pathogens possess remarkable genomic plasticity, allowing them to adapt to environmental perturbations and acquire antifungal resistance. Large-scale genomic alterations including aneuploidies, loss of heterozygosity (LOH), and chromosomal rearrangements can impact the expression of drug targets, efflux pumps and other factors that contribute to resistance. The emergence of aneuploidy provides a unique opportunity for cells to rapidly develop genotypic and phenotypic diversity without permanently committing to the mutant genotype. (Selmecki *et al.*, 2010; Chen *et al.*, 2012) The fungistatic azoles cause alterations in DNA content and cell morphology through an ordered series of aberrant cell cycle events that select for aneuploidy formation. (Harrison *et al.*, 2014) Karyotype variability conferring azole resistance has been extensively studied in *C. albicans* and is commonly observed in both resistant clinical isolates and laboratory strains. A specific aneuploidy on the left arm of chromosome 5 is the most prevalent. (Selmecki *et al.*, 2006; Coste *et al.*, 2007) This common segmental aneuploidy consists of an isochromosome composed of two identical left arms of Chr5 flanking a centromere [i(5L)]. Gain and loss of i(5L) tightly correlates with the gain and loss of azole resistance. The resistance associated with i(5L)-carrying isolates is mediated by increased copy numbers of ERG11 and TAC1, as well as gain-of-function mutations and LOH of these resistance determinants. (Selmecki *et al.*, 2008) There are forms of i(5L) that incorporate a region of chromosome 3 containing MRR1, allowing for further resistance to develop. (Selmecki *et al.*, 2009)

Additionally, aneuploidies of chromosomes 3, 4 and 6 have been implicated in azole resistance in *C. albicans*. (Hill *et al.*, 2013; Andersom *et al.*, 2017; Hirakawa *et al.*, 2017) Experimental evolution in the presence of azoles and the calcineurin inhibitor FK506 revealed extensive aneuploidies in strains that evolved resistance to the drug combination. (Hill *et al.*, 2013) The aneuploidy common in all resistant lineages was an increased copy number of chromosome 4, suggesting that this chromosome may harbour important resistance determinants (Hill *et al.*, 2013). Further, experimental evolution in a strain sensitised to azoles due to loss of the GTPase activator protein Rgd1 resulted in an amplification of both chromosome 7 and a large segment of chromosome 3, which accompanied suppression of the azole-sensitivity phenotype. Overexpression of a transporter gene on the affected region of chromosome 3 was sufficient to confer the suppression phenotype (Mount *et al.*, 2018).

LOH can also have a profound impact on drug resistance and is common in genomic regions containing determinants of azole susceptibility. LOH events can occur over short regions or over entire chromosomes. In *C. albicans,* LOH for the transcription factors TAC1 and MRR1 have been reported in azole-resistant isolates (Coste *et al.*, 2006; Dunkel *et al*, 2008). For both TAC1 and MRR1, LOH can generate two copies of a hyperactive allele, which increases the expression of genes encoding Cdr1 and Cdr2, or Mdr1, respectively (Coste *et al.*, 2006). Constitutive upregulation of ERG11 has also been reported as a consequence of LOH in the transcription factor UPC2 in *C. albicans* (Heilmann *et al.*, 2010). Interestingly, work has shown that a plethora of stress conditions increase mitotic recombination rates in *C. albicans*, leading to an increase in LOH rates (Forche *et al.*, 2011). This suggests a mechanism by which *C. albicans* can achieve greater genotypic diversity upon exposure to environmental perturbations, which can have important consequences given the lack of a conventional sexual cycle in this organism.

The incidence of echinocandin and polyene resistance arising from chromosomal abnormalities is infrequent compared to azole resistance in *C. albicans*. Nevertheless, rare genomic alterations have been reported in echinocandin-resistant isolates. Stepwise acquisition of micafungin resistance by homozygous hot-spot mutations in FKS1, followed by LOH, was identified as a mechanism of echinocandin resistance in *C. albicans* (Niimi *et al.*, 2010). Additionally, while an increase in chromosome 5 copy number confers azole resistance, a chromosome 5 monosomy causes decreases in β−1,3-glucan, increases in chitin, and decreases in echinocandin susceptibility (Yang *et al.*, 2013). Finally, trisomy of chromosome 2 reduces the susceptibility of *C. albicans* to caspofungin (Yang *et al.*, 2019). Adaptation to caspofungin in these strains is mediated by a calcineurin-mediated mechanism that is independent of the downstream transcriptional regulator Crz1 (Yang *et al.*, 2019).

2.5 Therapeutic Strategies to Combat Antifungal Resistance

Although advances are being made to improve our current antifungal arsenal and to identify novel drug targets, the rate at which antimicrobials are being developed is vastly outpaced by the rate at which resistance is emerging. Thus, examining alternative approaches, beyond targeting essential genes or cellular functions, for the development of new antifungal treatments is imperative to bolster the antifungal pipeline. A variety of strategies are in the early stages of being explored, including the use of combination therapy, the development of antivirulence agents and modulating host immune responses to fungal pathogens to combat or prevent infection **(**Spitzer *et al*., 2017).

2.5.1 Combination Therapy

Combination therapy is routinely implemented for the treatment of diverse infectious diseases, including malaria, tuberculosis and HIV/AIDS, while it's potential in antifungal treatment is only beginning to be explored. (Robbins *et al*. 2016; Spitzer *et al*., 2017) Specific drug combinations can overcome existing resistance and/or delay or prevent further resistance from emerging through numerous modalities. For instance, by combining multiple drugs with distinct mechanisms of action, the number of intracellular processes targeted increases, which makes evolution of resistance more difficult as the pathogen must acquire mutations in multiple genes (Robbins *et al*., 2017; Spitzer *et al*. 2017). Additionally, combination therapy can increase the fungicidal efficiency of normally fungistatic drugs (e.g. the azoles), reducing pathogen population sizes and thus, the probability of resistance arising (Onyewu *et al*., 2003; Spitzer *et al*., 2017). Importantly, in rare cases, a phenomenon known as selection inversion can occur, reversing drug resistance by neutralising selective advantages and imposing direct fitness costs on the evolution of drug resistance (Baym *et al*., 2016). Finally, combination therapy has the potential to unveil additional efficacious treatments given that fungal biology is controlled by highly interconnected and functionally redundant networks of biological interactions (Constanzo *et al*., 2010; Constanzo *et al*., 2011; Constanzo *et al*., 2016). The potential for targeting synthetic lethal interactions provides an expanded target space for efficacious combinations of compounds that on their own have little impact on fitness but combined are lethal. Together, this highlights that implementation of combination therapy is a beneficial avenue to address and treat drug-resistant fungal.

2.5.2 Targeting Virulence

Targeting fungal virulence provides a complementary approach to the development of fungicidal and fungistatic agents. In principle, these therapeutics simply aim to occlude the ability of a microbe to cause harm to its host. Although the potential utility of targeting nonessential genes has only recently been appreciated, there are already examples of antibiotics that target essential structures rather than essential proteins, as with daptomycin and the Gram-positive cell wall (Baltz, 2009). Targeting virulence factors offers many benefits including expanding the repertoire of antifungal targets, minimising effects on the host mycobiome and reducing selection pressure for the evolution of drug resistance (Clatworthy *et al*., 2007; Gauwerky *et al*., 2009).

2.5.3 Targeting Host Immunity

Modulating the host immune system by bolstering its capacity to resist fungal infection is another alternative therapeutic avenue to combat fungal infections. Development of preventative immunotherapeutics and vaccines against fungal pathogens has been explored with varying degrees of success. Making use of a shared fungal antigen present in multiple common pathogenic genera led researchers to focus on cell wall β-glucans for vaccine development (Armstrong-James *et al*., 2017). Although they are poorly immunogenic, conjugation of β-glucans to diphtheria toxoids has shown great promise in mounting strong anti-βF-glucan responses in murine models (Torosantucci *et al*., 2005). Conjugate β-glucan-based vaccines have demonstrated initial potential in restricting infection and prolonging survival in murine models of *Candida*, *Cryptococcus*, and *Aspergillus* infection, suggesting that they may present a valuable broad-spectrum therapeutic strategy following further development (Torosantucci *et al*., 2005; Bromuro *et al*., 2010; Pietrella *et al*., 2010; Specht *et al*., 2015).

The *Candida* agglutinin-like sequence (ALS) protein Als3, is a hyphal-specific glycophosphatidylinositol cell wall protein with multifunctional adhesin and invasion properties (Sui *et al*., 2017; Singh *et al*., 2019). Als3 is highly immunogenic, eliciting robust anti-Als3 B-cell responses and T-cell responses in mice, as well as anti-Als3 antibody responses in humans (Schmidt *et al*., 2012; Ibrahim *et al*., 2013). Early analysis of a recombinant N-terminal Als3 vaccine, NDV-3A, demonstrated protective efficacy in vaginal, oral and systemic murine models of *C. albicans* infection (Spellberg *et al*., 2006; Ibrahim *et al*., 2013). Subsequent progression through a phase II randomised clinical trial revealed the safety and efficacy of this vaccine in treating patients with recurrent vulvovaginal candidiasis (Edwards *et al*., 2018).

CHAPTER 3

METHODOLOGY

3.1 Sample Collection

Water samples from Mzingwane and Mtshabezi Dams were collected from different sites of the dams at different depths to make composite samples. 1 ml of each of the several samples was added to 9 ml of nutrient broth and incubated at 37 °C for 24 hours for enrichment. After 24 hours of incubation, the enriched samples were used to inoculate nutrient agar. 100 µl of each enriched sample was added to the surface of the agar using a micropipette and spread to cover the surface of the agar using a sterile glass spreader. The plates were incubated at 37°C for 24 hours. The plates were then observed for colony morphology. Colonies that were suspected to be that of *Candida albicans* were sub-cultured on Mueller Hinton Agar and then sub-cultured again on Sabouraud Dextrose Agar and Candida Ident Agar until pure isolates of *Candida albicans* obtained.

3.2 Fungal Identification

3.2.1 Morphological Identification

The morphological characteristics such as size, shape and colour were used in identifying *Candida albicans* on Candida Ident Agar and Sabouraud dextrose agar (SDA).

3.2.2 Microscopical Identification

3.2.2.1 Lactophenol Cotton Blue (LPCB) Assay

A suspension of *Candida albicans* cells was prepared in sterile water. 2 to 3 drops of the cell suspension were applied on a glass slide. 2 drops of lactophenol cotton blue solution was added to the slide and incubated at room temperature for 5 minutes. After incubation, the slide was viewed under a light microscope at x100.

3.2.2.2 Germ Tube Test

Clotted bovine blood was centrifuged at 5 000 g for 5 minutes to obtain serum. 1,5 ml of the serum supernatant was transferred to a 2 ml aliquot tube. A single colony of *Candida* *albicans* was suspended in the bovine serum. The suspension was incubated at 37°C for 2 hours. After incubation, 2 drops of the suspension were placed on a glass slide and viewed under a light microscope x100.

3.2.3 Chromogenic Agar Assay

Candida Ident Agar (Chromogenic Agar) imbued with chloramphenicol to inhibit bacterial growth was prepared by adding sterile distilled water to the agar powder and stirred. The solution was boiled until it was clear. The media was allowed to cool before it was poured onto agar plates. The media was inoculated with *Candida albicans* and incubated for 24 hours at 37 °C. Results were observed the following day.

3.3 Antimicrobial Sensitivity Tests

3.3.1 Kirby-Bauer Disc Diffusion Assay

Antifungal discs were placed on a lawn of *Candida albicans* seeded on the surface of Mueller Hinton agar using sterilised tweezers. Light pressure was applied to ensure full contact of the disc with the agar. The plates were incubated for 24 hours. Where there was a zone of inhibition, the diameter was measured and compared to a CLSI M-38A reference chart to determine the susceptibility of *Candida albicans* to each antifungal. CLSI (Clinical and Laboratory Standards Institutute) is a global leader in the development of medical laboratory standards and guidelines which aims to provide the most effective and efficient laboratory testing services and to continually improve laboratory quality and patient care. The CLSI M-38A document used for the Kirby-Bauer diffusion assay is this research is the 2022 version.

3.4 Molecular Work

3.4.1 Preparation of Samples for DNA Extraction

1 litre of LB broth was prepared by using the following constituents:

- Tryptone (10g)
- Yeast extract (5g)
- Sodium Chloride (10g)

Distilled water was added to the constituents, stirred until dissolved and autoclaved for 15 minutes at 121 °C. The broth was inoculated with *Candida albicans* and incubated at 37 °C for 72 hours.

3.4.2 DNA Extraction

The inoculated LB broth was poured into Epperndof tubes and centrifuged at 10 000 g for 5 minutes to obtain a sizeable pellet of *Candida albicans* cells. The broth supernatant was discarded and nuclease free water was added to the cells. The cells were treated with 30 µl of 10% SDS and incubated in a water bath at 70 °C for 30 minutes. 3 µl of 20 m/g Proteinase K was added to the cells and incubated in a water bath for 90 minutes at 37 °C. 500 µl solution of phenol-chloroform was added to the solution which was centrifuged at 10 000 g for 10 minutes to separate the solution into the organic phase and aqueous phase. The aqueous phase was transferred to sterile Epperndof tubes and treated with absolute ethanol and 5 M sodium chloride and incubated in a -20 °C freezer overnight. The samples were thawed and centrifuged at 13 000 g for 5 minutes to obtain a pellet. The pellet was treated with ice cold 70% ethanol, dislodged into solution and centrifuged at 13 000 g for 5 minutes. The supernatant was discarded and allowed to dry. 50 µl of nuclease free water was added to the DNA pellet and stored at -20 °C for further use in molecular work.

3.4.3: Polymerase Chain Reaction

The assay was performed in PCR tubes in a 25 µl reaction volume and run on an Applied Biosystems, ABI 2720 Thermal Cycler, 2007 model, USA. The reaction mixture contained 5µl of the reaction buffer (New England BioLabs, USA), 0.5 µl of deoxynucleotide triphosphate (dNTP) mix with concentration of 10 mM, 0.5 µl each of the forward and reverse primers (Inqaba Biotech, SA) with a concentration of 0.5 mM, 0.125 µl of One Taq DNA polymerase (New England BioLabs, USA), 1.0 µl of the template DNA, and this mixture was topped up to 25 µl by adding 17.375 µl of nuclease free water. The PCR reaction mixture was subjected to the following thermal cycling parameters/programme, denaturation at 94 ℃ for 2 minutes, 35 cycles (denaturation 94 ℃ for 30 seconds, annealing at 60℃ for 45 seconds, extension 68 ℃ for 1 minute) and then final extension at 72 ℃ for 7 minutes. 15 µl (10 µl amplicon and 5 µl loading dye (New England Biolabs, USA) of the amplified products were run on 1 % ethidium bromide-stained agarose gel along with the molecular weight marker (100bp DNA plus, a product of Fementas Incorporated, Vilnius, Lithuania).

Table 3.1: Primer Sequences for Amplified Genes

Gene	Primer Sequence	Amplicon Size
ERG11	F: ATGGGTATAATTATAGTTTTCCAT R: TCTGTACCATGAGCGGAAAC	1587 bp
ALS1	F: ATGTTTCCAATTGTTACTTTCATTG R: TTAAGAATACAACAACAACCAACCA	318 bp
MDR1	F: GCTGGTGGTTGGAGGTTATTG R: CCACCACCAGCAACAATAA	4500 bp

3.4.4 Gel Electrophoresis

1% agarose gel was prepared by dissolving 1 g of agarose powder in 100 ml TAE buffer and heated until the agarose powder dissolved. 10 µl of ethidium bromide was added to the agarose gel. The gel was poured into a tray with combs and allowed to solidify, upon which it was placed in TAE buffer where current was allowed to pass through after having loaded PCR products into the agarose gel wells. The current was allowed to run for 45 to 50 minutes. The gel was viewed under UV light to observe the gene bands.

CHAPTER 4

RESULTS

4.1 Microbial Analysis

A total of 20 isolates of *Candida albicans* were obtained from water samples collected from Mzingwane and Mtshabezi Dam. Confirmatory tests were performed for fungal identification of *Candida albicans*. The results are as follows in ***Table 4.1*** below.

Candida albicans was isolated and confirmed using a germ tube test, chromogenic media assay and lactophenol cotton blue assay. *Candida albicans* grows well on differential and selective media at an optimum temperature of 37 °C. For chromogenic assays, Candida Ident Agar was employed for identification of *Candida albicans* on primary isolation. Perry and Miller (1987) reported that *Candida albicans* produces an enzyme ß-N-acetylgalactosaminidase and according to Rousselle *et al.* (1994), incorporation of chromogenic or fluorogenic hexosaminidase substrates into the growth media help in identification of *Candida albicans* isolates directly on primary isolation. Candida Ident Agar is a selective and differential media which facilitates rapid isolation of yeasts from mixed cultures and allows differentiation of *Candida* species namely *Candida albicans*, *Candida krusei*, *Candida tropicalis* and *Candida glabrata* on the basis of colouration and colony morphology. On Candida Ident Agar, Candida albicans grows as light green to green smooth colonies.

Table 4.1: Cell and Colony Morphology/Presumptive Tests for Candida albicans

Morphology	***Candida albicans***
Growth	• *Candida albicans* grows well on differential, selective and general purpose media at an optimum temperature of 37 °C in 24 to 48 hours.
Colony	• On Candida Ident Agar, the colonies appeared as smooth colonies that are light green in colour. • On Sabouraud Dextrose Agar, the colonies appeared as white smooth colonies. • On Mueller Hinton Agar, colonies appeared as smooth, creamy and pasty.
Microscopy	• *Candida albicans* is a germ tube positive fungal yeast with oval or elliptical yeast cells of 3-5 µm in size, germ tubes protruding from the yeast cells that have a length between from 5-10 µm and hyhae that form as the germ tube grows, with a diameter of 2-4 µm. • On staining with lactophenol cotton blue stain, *Candida albicans* appears as blue oval shaped cells under a light microscope.

The morphological results on **Table 4.1** represent presumptive tests for our fungal isolates and they conform to the information from literature and are indicative of *Candida albicans.*

Table 4.2: Biochemical/Confimatory Tests for Candida albicans

Biochemical Tests	**Characteristics of *Candida albicans***
Germ tube test	Positive
Candida Ident Agar	Positive
Lactophenol cotton blue stain	Positive

4.2 Antimicrobial Susceptibility Tests

Figure 4.1 below summarises the resistance patterns of Candida albicans to fluconazole, itriconazole and posaconazole. For the 20 isolates of *Candida albicans* employed for this study, it was observed that all 20 isolates (100%) were resistant to fluconazole, 17 isolates (85%) were resistant to itriconazole and 10 isolates were resistant to posaconazole. 3 of the 20 isoaltes (15%) were susceptible to itriconazole and 10 of the isolates were susceptible to posaconazole. We noted that it was unsual for environmental isolates of *Candida albicans* to have such a high resitance to clinical antifungals unless if the water sources of Mzingwane and Mtshabezi Dams were contaminated by clinical effluent and waste water that was not treated or inadequatelt treated. It could also be an issue of cross-resistance due to exposure to agricultural azole fungicides, resulting in the mutation or overexpression of ERG11, whose alterations in its sequence results in resistance to clinical azoles by *Candida albicans*. Any of these suppositions could be a result of a high level of resistance to clinical azole used in this research.

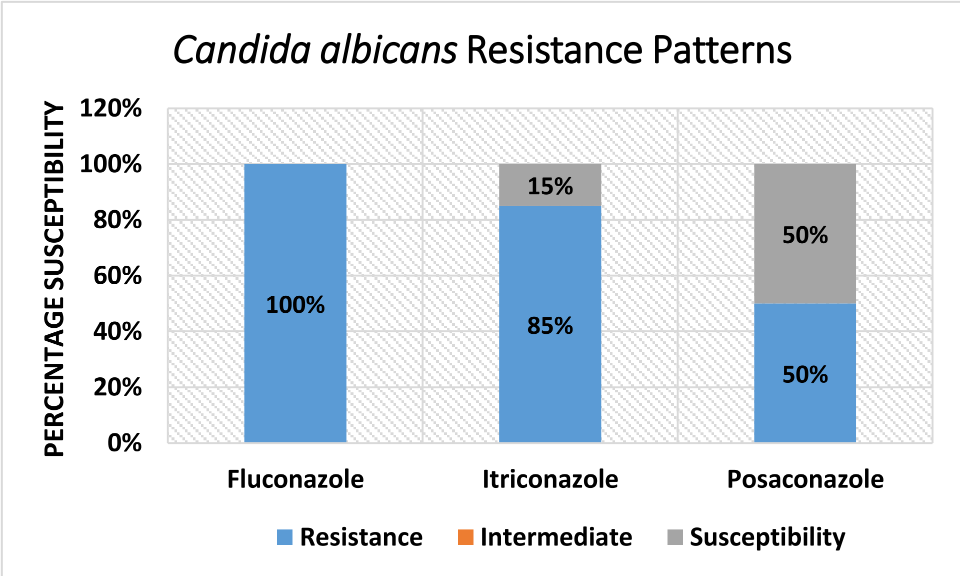


Figure 4.1: Candida albicans Susceptibility, Intermediate and Resistance Patterns (%), with it being most resistant to fluconazole and itriconazole and most susceptible to posaconazole.

4.3 Molecular Analysis Results

9 isolates that were resistant to all three antifungals used for sensitivity tests−fluconazole, itriconazole and posaconazole−were selected for molecular work. Overexpression and mutations of **ERG11** gene results in azole resistance particularly fluconazole. Overexpression of the **MDR1** gene encodes a multi-drug efflux pump, resulting in multidrug resistance in *Candida albicans*. The **ALS1** gene is a virulence gene which encodes glycoproteins involved in pathogenesis, aggregation and adhesion of *Candida albicans* to mucosa and epithelial cells.

4.3.1 PCR-based Detection of ERG11 Gene

PCR amplification of the ERG11 gene gave positive signals for the 9 isolates selected for amplification of the gene as indicated on the electrophoretogram. (***see Figure 4.2***) Our results, which are genotypic in nature, were suggestive of the presence of ERG11 gene and its expression as indirectly confirmed by the pronounced antimicrobial resistance of *Candida albcans* isolates to fluconazole, itriconazole and posaconazole in the phenotypic analysis employed in our study.

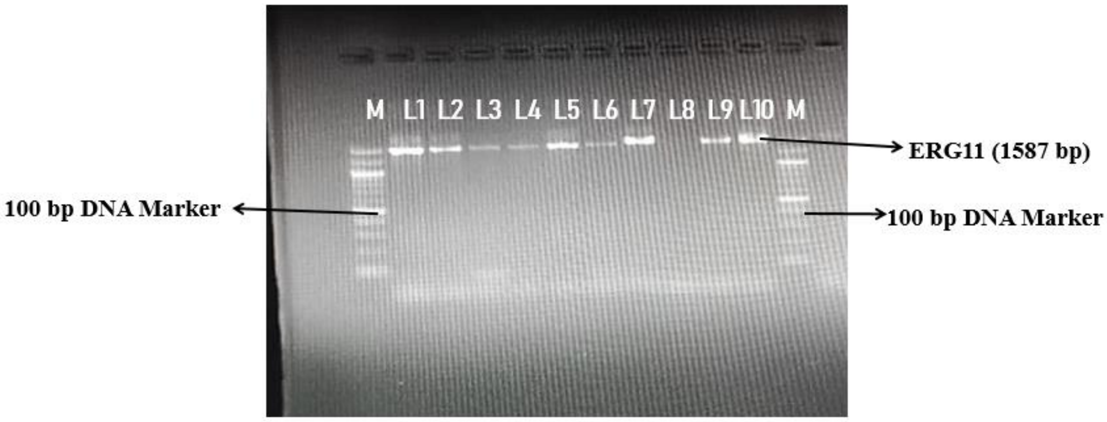


Figure 4.2: Electrophoretogram of PCR amplification product of suspected ERG11 target gene. ERG11 is 1587 bp long and mutation and overexpression of the gene results in C. albicans developing resistance to azole antifungals.

4.3.2 PCR-based Detection of ALS1 Gene

PCR amplification of the virulence factor gene ALS1 gave positive signals as shown on the electrophoretogram for 5 out of 6 of the isolates selected for ALS1 gene amplification. (***see Figure 4.3***) Our results, which are genotypic in nature, were suggestive of the presence of the ALS1 gene which encodes a Als1 glycoprotein for adhesive and aggregative interactions. The ALS1 gene is also implicated in the cell size homeostasis of *C. albicans*.

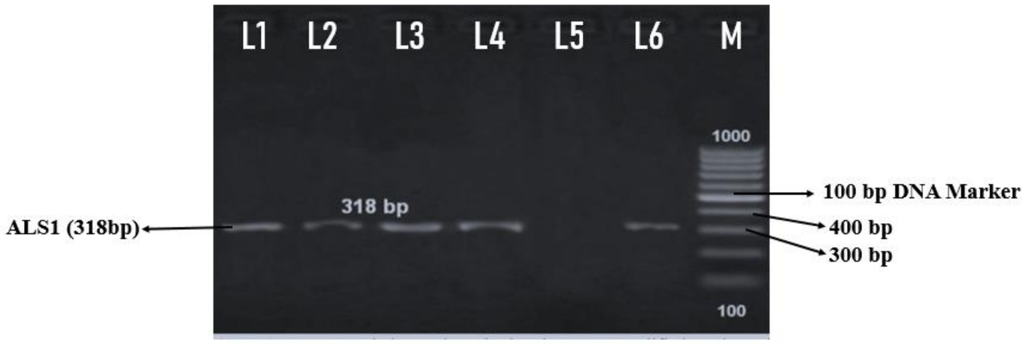


Figure 4.3: Electrophoretogram of PCR amplification product of suspected ALS1 target gene. ALS1 is a gene which encodes the Als1 glycoprotein responsible for adhesive and aggregative interactions.

4.3.3 PCR-based Detection of MDR1 Gene

PCR amplification of the multidrug resistance factor, MDR1 gave positive signals for 7 of the 9 isolates selected for MDR1 gene amplification as shown on the electrophoretogram (***see Figure 4.4***). Our results, which are genotypic in nature, were suggestive of the presence of the gene for multidrug resistance and its expression.

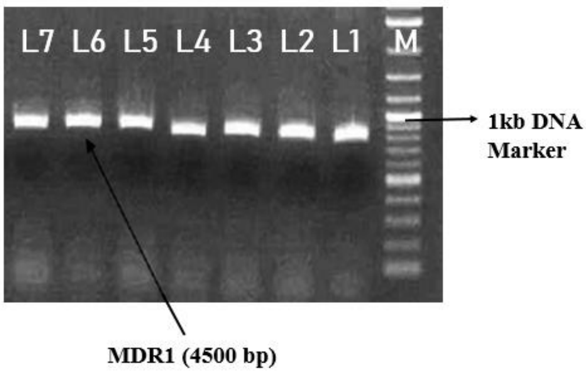


Figure 4.4: Electrophoretogram of PCR amplification product of suspected MDR1 target gene. MDR1 encodes a multidrug efflux pump of the major facilitator superfamily, which is responsible for pimping out azole antifungals, cellular waste products and other toxic compounds.

4.4 PCR Optimization

PCR reaction conditions were optimized to annealing temperatures ranging from 53 °C to 58 °C, denaturing temperatures ranging from 94 °C to 95 °C and an extension temperature of 72 °C. A 10 µl PCR reaction was performed. 5 µl of the enzyme master mix was used, with 0,16 µl of each of the forward and reverse primers, 2,68 µl of nuclease free water and 2 µl of DNA. The number of cycles ranged from 30 to 35. This was done so as to increase the probability of primers annealing to the target sequence and gene amplification and hence, the efficiency of PCR.

CHAPTER 5

DISCUSSION

Candida albicans was isolated from environmental water samples collected from Mzingwane and Mtshabezi Dams with the study being conducted for a duration of 4 months. This research was in two parts; the phenotypic (microbiological) part and the genotypic (molecular) part. The observed microbial growth and isolation of *Candida albicans* from Mzingwane and Mtshabezi Dams is indicative of the growth of microorganisms in water bodies and this is attributed to a number of possible contaminations from the environment, contamination from animal waste and contamination from human activities. *Candida albicans* is ubiquitous yeast that can naturally be present in the environment, such as in soil, decaying organic matter or even on the skin of animals. Environmental changes can lead to the release of these yeasts into water bodies. All of the aforementioned reasons attribute to the presence of *Candida albicans* in Mzingwane and Mtshabezi Dams. According to Chaieb and co-workers (2011), *Candida albicans* can endure long periods of starvation in water. From their research, Chaieb *et al*., (2011) showed that *Candida albicans* cells remain viable and cultivatable within a lengthy period followed by a decline in cultivability and a drop of plate counts to less than 20 cells/ml after 150 days in tap water, 190 days in rain water and 200 days in seawater. They also observed that the number of viable cells dropped to 10% after 150 days in seawater, 180 days in rain water and 210 days in tap water (Chaieb *et al*., 2011). This indicates that *C. albicans* can survive in an environment that is deprived of its nutritional requirements and despite the drop in the number of viable cells, the cells still remain cultivatable and hence can survive in different aquatic environments for a long time, fresh water bodies included.

The presence of antifungal resistant *Candia albicans* isolates from Mzingwane and Mtshabezi Dams can be due to various reasons. The first one being that fungal-to-fungal interactions in a niche can lead to the development of adaptive traits that increase the chances of survival against other fungal species they are competing with for space and resources. For example, studies have shown that *Candida albicans* can develop fluconazole resistance when co-cultured with *Candida glabrata* or *Saccharomyces cerevisiae*. *Aspergillus fumigatus* can develop azole resistance when co-cultured with *Aspergillus terreus*. Fungal-to-fungal interactions can lead to the transfer of resistance genes between species, promoting the spread of antifungal resistance by means of horizontal gene transfer, upregulation of efflux pumps, alterations in cell wall composition and increased biofilm formation (Tackling the emerging threat of antifungal resistance to human health, 2022). The environmental strains of *Candida albicans* having such a high incidence of resistance could have been a result of contamination of fresh water catchment areas by:

- Untreated or inadequately treated wastewater effluent from hospitals, clinics and households which can contain clinical samples of *C. albicans*. The effluent can enter freshwater bodies through sewage overflows or effluent discharge.
- Recreational human activities can also pose a threat of introducing *C. albicans* into fresh water bodies.
- Urban runoff from urban areas can transport *C. albicans* from contaminated surfaces, drains, and sewage systems.
- Animal carriers like birds, bats, or other wildlife might have *C. albicans* in their gastrointestinal tract, which they can pass off into fresh water bodies through faecal matter.

All these suppositions are trying to explain why there was a high rate of antifungal resistance in *C. albicans* isolated from Mzingwane Dam and Mtshabezi Dam. The reasoning is that there is a possibility that not all isolates of *C. albicans* used in this study were originally environmental isolates. The dams, with the possibility of contamination by means listed above, could contain clinical isolates from the sources of contamination, prior to which they might have already been exposed to clinical azoles and developed resistance, hence their resistance when tested against fluconazole, itriconazole and posaconazole.

However, studies have noted a growing concern in antifungal resistance in clinical *C. albicans* isolated from patients that have never had prior antifungal treatment. Bastos and co-workers (2021) posited that this could be due a development of cross-resistance in environmental species of *Candida*. In their study, Bastos and co-workers (2021) noted that although in-host resistance acquisition is the main route of azole resistance development in *Candida* spp., the isolation of azole-resistant strains from patients with no prior history of antifungal treatment has become common. One of the explanations for this phenomenon may be in the environment. It was supposed that *Candida* spp. found in the environment may acquire resistance in that place due to fungicides, particularly environmental azoles, which could be the stressor-selecting pressure, resulting in cross-resistance. It has been shown that the fluconazole MIC values are higher in *Candida* species isolated from the surface of nonorganic fruits (sprayed with fungicides) compared to those collected from organic ones (without agrochemical). Cross-resistance can also be achieved in vitro by exposing yeasts to agricultural azoles. Fluconazole and posaconazole resistance, for example, were selected in *C. glabrata* after a previous exposure to the fungicide prochloraz. In addition, susceptible *C. parapsilosis* species became more resistant to fluconazole, itraconazole and voriconazole after being cultured in a medium supplemented with the fungicides tetraconazole and tebuconazle (Bastos *et al*., 2021). This shows that fungicides may be inducing resistance to clinical azoles in *Candida* spp. mainly through activation of overexpression of efflux pumps and ERG11 genes. They are also affecting its morphophysiology; however, it is unclear, if those alterations impact *Candida* virulence. (Bastos *et al*., 2021)

In this study, 100%, 85% and 50% of the *C. albicans* isolates were resistant to fluconazole, itriconazole and posaconazole respectively. In a similar study by Yeniserhirli *et al*., (2015) in Tokat, Turkey, 34%, 21% and 14% of *Candida albicans* isolates were resistant to fluconazole, itriconazole and posaconazole. In their study, they noted that the widespread use of fluconazole, due to its relative safety and high oral bioavailability for treatment and prophylaxis may be the cause of high azole resistance rates observed. In addition, decreased susceptibility to fluconazole is associated with decreased susceptibility to other azoles (Yeniserhirli *et al*., 2015).

Fluconazole resistance is conferred by a mutated ERG11 gene and also by overexpression of ERG11 due to gain-of-function in transcription factors MRR1 and TAC1. MRR1 regulates the expression of MDR1 and TAC1 regulates the expression of CDR1 and CDR2 (Popp *et al*., 2019). *C. albicans* was known to be an asexual microorganism until the sequenced genome revealed that *C. albicans* possesses a mating-type locus (MTL) similar to that of fungi with a known sexual cycle. If *C. albicans* is MTL-heterozygous, possessing both MTLα and MTLa alleles, it is inhibited from mating because presence of both alleles leads to the formation of a heterodimeric repressor complex. This complex binds to the MTL and represses the expression of genes required for mating, effectively hindering the mating process. This is a regulatory mechanism that ensures that only cells with a single mating type, either MTLα or MTLa, can mate, maintaining genetic integrity and preventing aberrant sexual reproduction (Popp *et al*., 2019). MTL-heterozygous *C. albicans* can achieve homozygosity through genome rearrangements and mitotic recombination. It can then switch to be mating-competent, combining their adaptive traits, increasing their survival as is the case with fluconazole resistance. And so, in an originally clonal population, cells that acquired resistance mutations and have become homozygous for the mutated allele are likely to become mating-competent and can share their resistance mechanisms by parasexual recombination to produce highly drug resistant cells (Popp *et al*., 2019).

In our study, isolates selected for gene amplification were all resistant to fluconazole, itriconazole and posaconazle and all these were positive for ERG11 after gene amplification using conventional PCR. ERG11 gene encodes lanosterol 14α-demethylase, the target of the azole antifungals. Lanosterol 14α-demethylase is involved in the synthesis of sterols which function to maintain fluidity in eukaryotic membranes. The azole class of antifungals inhibits ergosterol biosynthesis and allows for the accumulation of toxic methylated sterol precursors, which affects the integrity of the cell membrane (Flowers *et al*., 2015). But due to overuse of the azole antifungal class, as the only available option for systematic antifungal treatment and available option for prophylaxis and therapy, *C. albicans* has adapted to being resistant to azoles. Mutations in ERG11 that result in an amino acid substitution that alter the abilities of the azoles to bind to and inhibit ERG11, resulting in resistance (Flowers *et al*., 2015).

From ***Table 4.1***, all isolates of *Candida albicans* appeared as large, smooth colonies and green in colour on Candida Ident Agar, on Sabouraud Dextrose Agar, isolates appeared as smooth white colonies and Mueller Hinton Agar, colonies appeared as smooth, creamy and pasty. *Candida albicans* is a germ tube positive yeast, with oval or elliptical yeast cells with hyphae that form as the germ tube grows. A positive germ tube test on its own should be a confirmatory test for *Candida albicans*, but additional tests had to be done since *Candida albicans* is not the only germ tube positive species of the *Candida* genus. According to Calderone (2001), *Candida albicans* and *Candida duliniensis* are polymorphic due to their ability to form hyhae/pseudohyphae and both are referred to as germ tube positive. And hence to avoid misidentification of another *Candida* species, as *Candida albicans*, a chromogenic agar assay was performed. Candida Ident Agar was used for this assay. Perry and Miller (1987) reported that *Candida albicans* produces an enzyme ß-N-acetylgalactosaminidase and according to Rousselle *et al.* (1994), incorporation of chromogenic or fluorogenic hexosaminidase substrates into the growth media help in identification of *Candida albicans* isolates directly on primary isolation. Candida Ident Agar is a selective and differential media which facilitates rapid isolation of yeasts from mixed cultures and allows differentiation of *Candida* species namely *Candida albicans*, *Candida krusei*, *Candida tropicalis* and *Candida glabrata* on the basis of colouration and colony morphology. *Candida albicans* grows as light green to green colonies on Candida Ident Agar.

The detection of the ERG11, MDR1 and ALS1 genes in environmental isolates of *Candida albicans* is important for several reasons. Firstly, it can help to guide appropriate antifungal treatment, as infections caused by the genus *Candida,* in particular *Candida albicans* that carry these genes may be resistant to a broad spectrum of classes of antifungals. Secondly, it can inform infection control measures, as these fungi, particularly *Candida albicans* and *Candida auris* may be more likely to spread in healthcare settings. Finally, the monitoring of ERG11 and ALS1 genes prevalence over time can help to track the emergence and spread of antifungal resistance in fungi of the *Candida* populations, which is important for public health planning and policy. It is worth noting that the detection of these genes in fungal isolates does not necessarily indicate that the fungi are causing an infection or that the patient is symptomatic. However, it is an important tool for surveillance and monitoring of antifungal resistance in fungal populations, which can help to guide appropriate treatment and infection control measures.

The findings in this study reflected greater susceptibility to posaconazole (50%) followed by itriconazole (15%), which might be due to the minimal use of such drugs for treatment. On the contrary, the high degree of resistance to fluconazole might be due to over-dependence on such antifungals. To note also, the widespread use of fluconazole, due to its relative safety and high oral bioavailability for treatment and prophylaxis may be the cause of high fluconazole resistance rates observed. The overall antibiogram of *Candida albicans* in this study showed a decreased susceptibility against all three azole antifungals. This might be due to the haphazard use of the drug in hospital settings and the availability of the drug as over-the-counter medicine is some settings, resulting in its misuse and abuse.

In this study, a PCR-based assay of ALS1 was done and 5 out of 6 isolates were found to be positive of the ALS1 gene. ALS (Agglutini-like sequence) genes are a cluster of genes involved in the pathogenesis and adhesion of *Candida albicans* to mucosa and epithelial cells in the vagina. According to a study similar to ours by Roudbarmohammadi *et al*., (2016), *Candida albicans* isolates with expression of at least one ALS gene were resistant to fluconazole and those without ALS gene expression were absolutely sensitive to fluconazole. This alludes to the fact that there is a correlation between fluconazole resistance and ALS1 gene expression. Since all of *Candida albicans* isolates expressing ALS1 genes were resistant to fluconazole, it seems reasonable to conclude that biofilm formation, conferred by the expression of ALS1, contributes to fluconazole resistance in *C. albicans* isolates. This finding tallies with the results of other studies according to which ALS1 was introduced as the major over-expressed gene in biofilm formation and adherence which in turn play an important role in drug inhibition to gain access to the fungal microcolonies Roudbarmohammadi *et al*., 2016).

CHAPTER 6

6.1 CONCLUSION

Environmental isolates were used for this study and it was observed that they have a high resistance to fluconazole, itriconazole and posaconazole. This can be explained by the use of fungicides in the environment, which resulted in cross-resistance, increasing resistance to clinical azoles. Although the effects of fungicides to virulence of *Candida* spp. is still unclear. Based on that, the use of azoles in human medicine and the environment requires surveillance and restrictions to minimize the risk of inducing clinical azole resistance in *Candida albicans*. However, it was supposed that the water catchment areas, that is Mazingwane Dam and Mtshabezi Dam, might have been contaminated by clinical isolates of *C. albicans*. The sources of contamination might include untreated or inadequately treated wastewater effluent from hospitals, urban runoff from sewage, recreational human activities and animals that are carriers of the fungi on fur, feathers and gastrointestinal tract. The isolates used in this study were positive for the presence of ERG11. For environmental isolates, this could have been caused by exposure to fungicides, parasexual recombination and competitive fungal-to-fungal interactions in a defined niche that resulted in the mutation and/or overexpression of ERG11 which confers resistance to azoles, in particular, fluconazole. Although the sample size used for this study was small, the results obtained were satisfactory and demonstrated that environmental pathogens can cause infections. This was largely significant as it indicated that if the exploration was to be expanded with a larger sample size, further of these virulence and antifungal resistance genes including the ones not conducted in this study could be identified.

6.2 RECOMMENDATIONS

A study on inhibitory effects of *Lactobacillus* species on *Candida* spp. would be beneficial in coming up with alternative ways for antifungal therapy and treatment due to the incidence of increased antifungal drug resistance. Due to the high incidence of fungal infections caused by *Candida* species and their increasing resistance to antifungal treatments, alternative therapies such as probiotics have been studied. It has been shown that several species of the genus *Lactobacillus* have anti-*Candida* activity, probably by direct inhibition, through competition for adhesion sites or production of secondary metabolites, and by indirect inhibition, through stimulation of the immune system of their host. (Ribeiro *et al*., 2019)

In a study by Zangl *et al*. (2019), it was observed that lactic acid has inhibitory effects against *Candida* spp. However, lactic acid production in *Lactobacillus* is strain specific as it was observed that no inhibition of *C. albicans* or *C. glabrata* was observed with the lactic acid concentration reached with the supernatant of a cultured *L. rhamnosus*. It was proposed that only elevated levels of lactic acid efficiently inhibit fungal growth. In support of this, supernatants of *L. rhamnosus*, *L. casei* and *L. acidophilus* exhibit an antifungal effect against *Candida* spp. only if harvested after prolonged incubation (24 hours to 48 hours) in which lactic acid would have accumulated in the medium (Zangl *et al*., 2019). Vilela *et al*. (2015) demonstrated the inhibitory effects of *L. acidophilus* on biofilm formation and filamentation. They determined that 24 hours was the best growth phase of the *L. acidophilus* culture to be able to inhibit *C. albicans* cells in *in vitro* biofilms, with 57% inhibition. They also analysed the effects of culture filtrate of *Lactobacillus* after 24 hours of growth on biofilms and there was a reduction in *C. albicans* cell growth by 45%.

A regular surveillance of antifungal resistance in *Candia albicans* from the environment and hospitals and their virulence profiles with a bigger sample size and different sampling sites is recommended to obtain more accurate results. A comparative study should be done for the following reasons:

- To investigate if environmental isolates have adapted to their environment, leading to changes in antifungal susceptibility and virulence factor expression.
- To identify potential environmental reservoirs of *C. albicans* that could lead to human infections.
- Understand the possible transmission routes between environmental and human hosts.

Studies can further be done for the detection of a wide variety of virulence genes and antifungal resistance genes, which will be beneficial in understanding pathogenicity of *Candida albicans* and assist in coming up with better treatment options for patients. Besides azoles, susceptibility and resistance patterns of other classes of antifungals, polyenes and echinocandins, must be studied to further understand the demography of antifungal resistance on a broader spectrum. This can include studying the patterns of resistance across different areas and populations, how widespread the resistance is, the characteristics of resistant strains and how resistance emerges, spreads and evolves overtime.

CHAPTER 7

REFERENCES

Anaissie E. J. (1992). Opportunistic mycosis in the immunocompromissed host: Experience at a cancer centre and review. Clin. Infect. Dis. 14: 43-53.

Aslam B., Wang W., Arshad M. I., Khurshid M., Muzammil S. and Rasool M. H. (2018). Antibiotic resistance: a rundown of a global crisis. Infect Drug Resist. 11: 1645-1658.

Banerjee S. N., Emori T. J. and Culver D. H. (1991). Secular trends in nosocomial primary bloodstream infections in the United States. 1980-1989 National Nosocomial Infections Surveillance System. Am. J. Med. 91(3B): 86S-89S.

Bennett J. E. (1996). *Antimicrobial agents: antifungal agents*. In: Goodman and Gilman's the Pharmacological Basis of Therapeutics, Gilman A. G., Rall, T. W., Sies A. S. and Taylor P. (Eds.). Pergamon Press, Elmsford, NY. p. 1165-1181.

Bhattacharya S., Sae-Tia S. and Fries B. C. (2020). Candidiasis and Mechanisms of Antifungal Resistance. 9 (312): 1-2.

Bongomin F., Gago S., Oladele R. and Denning D. (2017). Global and multinational prevalence of fungal diseases estimate precision. J Fungi. 3: 57.

Caceres D. H., Forsberg K., Welsh R. M., Sexton D. J., Lockhart S. R. and Jackson B. R. (2019). *Candida auris*: A review of recommendations for detection and control in health care settings. J Fungi. 5: 111.

Corner B. E. and Magee P. T. (1997). *Candida* pathogenesis: Unravelling the threads of infection. Curr. Biol. 7: R691-R694.

Correia A., Sampaio P., Vilanova M. and Pais C. (2015). *Candida albicans: Clinical relevance, pathogenesis and host immunity*. In: Human emerging and re-emerging infections: viral and parasitic infections, Sing S. K (Ed.). John Wiley and Sons, New Jersey. 1: 926-952.

Cutler J. E. (1991). Putative virulence factors of *Candida albicans*. Annu. Rev. Microbiol. 45: 187-218.

Dadar M., Tiwari R., Karthik K., Chakraborty S., Shahali Y. and Dhama K. (2018). *Candida albicans*-biology, molecular characterisation, pathogenicity and advances in diagnosis and control-an update. Microb Pathog. 117: 128-138.

De Rosa F. G., Garazzino S., Pasero D. C. and Peri G. D. (2009). Invasive candidiasis and candidemia: new guidelines. Minerva Anaestesiologica. 75: 453-458.

Dhasarathan P., AlSalhi M. S., Devanesan S., Subbiah J., Ranjitsingh A. J. A., Binsalah M and Alfuraydi A. A. (2021). Drug resistance in *Candida albicans* isolates and related changes in the structural domain of Mdr1 prtein. Journal of Infection and Public Health. p. 1848 – 1853.

Dupont B. (1992). *Antifungal therapy in AIDS patients*. In: New Strategies in Fungal Disease, Bennett J. E., Hay R. J. and Peterson P. K. (Eds.). Churchill Livingstone, London. p. 290-300.

Fox J. L. (1993). Fungal infection rates are increasing. ASM News. 10: 515-518.

Georgopapadakou N. H. and Walsh T. J. (1996). Antifungal agents: chemotherapeutic targets and immunological strategies. Antimicrob. Agents Chemother. 40: 279-291.

Ingham C. J., Boonstra S., Levels S., de Lange M., Meis J. F. and Schneeberger P. M. (2012). Rapid susceptibility testing and microcolony analysis of *Candida* spp. cultured and imaged on porous aluminium oxide. PLoS ONE. 7: e33818.

Kenafani Z. A. and Perfect J. R. (2024). Resistance to Antifungal Agents: Mechanisms and Clinical Impact. Antimicrobial Resistance. 2008: 46.

Kobayashi G. S. and Cutler J. E. (1998). *Candida albicans* hyphal formation and virulence: Is there a clearly defined role? Trends Microbiol. 6: 92-94.

Kornitzer D. (2019). Regulation of *Candida albicans* hyphal morphogenesis by endogenous signals. J Fungi. 5: 21.

Limon J. J., Skalski J. H. and Underhill D. M. (2017). Commensal fungi in health and disease. Cell Host Microbes. 22: 156-165.

Magee P. T. (1998). Which came first: the hypha or the yeast? Science 277: 52-53.

Negri M., Faria M., Guilhermetti E., Alves A., Paula C. and Svidzinski T. (2010). Hemolytic activity and production of germ tubes related to pathogenic potential of clinical isolates of *Candida albicans*. J Basic Appl Pharm. 31: 89-93.

Odds F. C. (1988). *Candida* and Candidosis. Bailliére Tindall, London.

Odds F. C. (1994). *Candida* species and virulence. ASM News 60: 313-318.

Perez J. C. and Johnson A. D. (2013). Regulatory circuits that enable proliferation of the fungus *Candida albicans* in a mammalian host. PLoS Pathogen. 9(12): e1003780.

Perfect J. R. (1996). Fungal virulence genes as targets for antifungal chemotherapy. Antimicrob. Agents Chemother. 40: 1577-1583.

Rex J. H., Rinaldi M. G. and Pfaller M. A. (1995). Resistance of *Candida* species to Fluconazole. Antimicrob. Agents Chemother. 1: 1-8.

Ryley J. F. and Ryley N. G. (1990) *Candida albicans* – do mycelia matter? J. Med. Vet. Mycol. 28: 225-239.

Sanglard D., Kuchler K., Ischer F., Pagani J. L., Monod M. and Bille J. (1995). Mechanisms of resistance to azole antifungal agents in *Candida albicans* isolates from AIDS patients involve specific multidrug transporters. Antimicrob. Agents Chemother. 39: 2378-2386.

Silva S., Negri M., Henriques M., Oliveira R., Williams D. W. and Azeredo J. (2011). Adherence and biofilm formation of non-*albicans Candida* species. Trends Microbio. 19: 241-247.

van den Bossche H., Dromer F., Improvisi I., Lozano-Chiu M., Rex J.H. and Sanglard D. (1998). Antifungal drug resistance in pathogenic fungi. Med. Mycol. 36 (1): 119-128.

Vincent J. L., Rello J., Marshall J., Silva E., Anzueto A. and Martin C. D. (2009). EPIC II Group of Investigators. International study of the prevalence and outcomes of infection in intensive care units. JAMA. 302: 2323-2329.

Wisplinghoff H., Seifert H., Tallent S. M., Bischoff T., Wenzel R. P. and Edmond M. B. (2003). Nosocomial bloodstream infections in paediatric patients in United States hospitals: Epidemiology, clinical features and susceptibilities. Pediatr Infect Dis J. 22: 686-691.

Achkar J. M. and Fries B. C. (2010). *Candida* infections of the genitourinary tract. Clin Microbiol Rev. 23(2): 253–273.

Akpan A. and Morgan R. (2002). Oral candidiasis. Postgrad Med J. 78(922): 455–459.

Albrecht A., Felk A., Pichova I., Naglik J. R., Schaller M. and de Groot P. (2006). Glycosylphosphatidylinositol–anchored proteases of *Candida albicans* target proteins necessary for both cellular processes and host–pathogen interactions. J Biol Chem. 281: 688–694.

Albuquerque P. and Casadevall A. (2012). Quorum sensing in fungi–a review. Med Mycol. 50: 337–45.

Allert S., Förster T. M., Svensson C. M., Richardson J. P., Pawlik T., Hebecke, B., Rudolphi S., Juraschitz M., Schaller M. and Blagojevic M. (2018). *Candida albicans*–induced epithelial damage mediates translocation through intestinal barriers. MBio. p. 9.

Almeida R. S., Wilson D. and Hube B. (2009). *Candida albicans* iron acquisition within the host. FEMS Yeast Res. 9: 1000–1012.

Alsomali M. I., Arnold M. A., Frankel W. L., Graham R. P., Hart P. A., Lam–Himlin D. M., Naini B. V., Voltaggio L. and Arnold C. A. (2017). Challenges to "classic" oesophageal candidiasis: Looks are usually deceiving. Am. J. Clin. Pathol. 147: 33–42.

Anderson M. Z., Saha A., Haseeb A. and Bennett R. J. (2017). A Chromosome 4 Trisomy Contributes to Increased Fluconazole Resistance in a Clinical Isolate of *Candida albicans*. Microbiology.163: 856–865.

Andriole V. T. (2000). Current and future antifungal therapy: New targets for antifungal therapy. Int. J. Antimicrob. Agents. 16: 317–321.

Ang B. S. P., Telenti A., King B., Steckelberg J. M. and Wilson W. R. (1993). Candidemia from a urinary tract source: Microbiological aspects and clinical significance. Clin. Infect. Dis. 17: 662–666.

Armstrong-James D., Brown G. D., Netea M. G., Zelante T., Gresnigt M. S., van de Veerdonk F. L. and Levitz S. M. (2017). Immunotherapeutic Approaches to Treatment of Fungal Diseases. Lancet Infect. Dis. 17: E393–E402.

Author C., Shokohi T., Reza Aghili S., Ali Boroumand M., Hashemi Fesharaki S. and Salmanian B. (2015). Intravenous Catheter–Associated Candidemia due to *Candida membranaefaciens*: The First Iranian Case. J. Tehran Univ. Heart Center. 10: 101–105.

Baddley J. W., Benjamin D. K., Patel M., Miró J., Athan E., Barsic B., Bouza E., Clara L., Elliott T. and Kanafani Z. (2008). *Candida* infective endocarditis. Eur. J. Clin. Microbiol. Infect. Dis. 27: 519–529.

Baltz R. H. (2009). Daptomycin: Mechanisms of Action and Resistance, and Biosynthetic Engineering. Curr. Opin. Chem. Biol. 13: 144–151.

Barton M., Shen A., O'Brien K., Robinson J. L., Davies H. D., Simpson K., Asztalos E., Langley J., Le Saux N. and Sauve, R. (2017). Early–onset invasive candidiasis in extremely low birth weight infants: Perinatal acquisition predicts poor outcome. Clin. Infect. Dis. 64: 921–927.

Basmaciyan L., Bon F., Paradis T., Lapaquette P., Dalle F. (2019). *Candida Albicans* Interactions with the Host: Crossing The Intestinal Epithelial Barrier. Tissue Barriers. 7: 1612661.

Basso V., d'Enfert C., Znaidi S. and Bachellier–Bassi S. (2019). *From genes to networks: The regulatory circuitry controlling Candida albicans morphogenesis*. In: Fungal Physiology and Immunopathogenesis: Current Topics in Microbiology and Immunology, Rodrigues M. (Ed.). Springer: Berlin/Heidelberg, Germany. 422: 61–99.

Baym M., Stone L. K. and Kishony R. (2016). Multidrug Evolutionary Strategies to Reverse Antibiotic Resistance. Science. 351: aad3292.

Beardsley J., Halliday C. L., Chen S. C. A. and Sorrell T. C. (2018). Responding to the emergence of antifungal drug resistance: Perspectives from the bench and the bedside. Future Microbiol. 13: 1175–1191.

Benito–Cruz B., Aranda–Romo S., López–Esqueda F. J., de la Rosa–García E., Rosas–Hernández R. and Sánchez–Vargas L. O. (2016). Oral *Candida* isolates and fluconazole susceptibility patterns in older Mexican women. Arch Gerontol Geriatr. 65: 204–210.

Brand A. and Gow N. A. (2009). Mechanisms of hypha orientation of fungi. Curr Opin Microbiology. 12: 350–7.

Brand A., Shanks S., Duncan V. M., Yang M., Mackenzie K. and Gow N. A. (2007). Hyphal orientation of *Candida albicans* is regulated by a calcium–dependent mechanism. Curr Biol. 17: 347–352.

Brand A., Vacharaksa A., Bendel C., Norton J., Haynes P. and Henry–Stanley M. (2008). An internal polarity landmark is important for externally induced hyphal behaviours in *Candida albicans*. Eukaryot Cell. 7: 712–720.

Brock M. (2009). Fungal metabolism in host niches. Curr Opin Microbiology. 12:371–376.

Bromuro C., Romano M., Chiani P., Berti F., Tontini M., Proietti D., Mori E., Torosantucci A., Costantino P. and Rappuoli R. (2010). Beta-Glucan-CRM197 Conjugates as Candidates Antifungal Vaccines. Vaccine. 28: 2615–2623.

Brown A. J. P., Haynes K., Gow N. A. R. and Quinn J. (2012). *Stress Responses in Candida*. In: *Candida* and Candidiasis. Calderone R. A. and Clancy C. J. (Eds.). ASM Press, Washington, DC. p. 225–242.

Brown G. D., Denning D. W. and Levitz S. M. (2012). Tackling human fungal infections. Science. 336: 647.

Bruno V. M. and Mitchell A. P. (2005). Regulation of Azole Drug Susceptibility by *Candida albicans* Protein Kinase CK2. Mol. Microbiol. 56: 559–573.

Calandra T., Roberts J. A., Antonelli M., Bassetti M. and Vincent J. L. (2016). Diagnosis and management of invasive candidiasis in the ICU: An updated approach to an old enemy. Crit. Care. 20: 125.

Cannon R. D, Holmes A. R, Mason A. B. and Monk B. C. (1995). Oral *Candida*: Clearance, colonisation, or candidiasis? Journal of Dental Research. 74: 1152–1161.

Cannon R. D., Lamping E., Holmes A. R., Niimi K., Baret P. V., Keniya M. V., Tanabe K., Niimi M., Goffeau A. and Monk B. C. (2009). Efflux–Mediated Antifungal Drug Resistance. Clin. Microbiol. Rev. 22: 291–321.

Carmon M., Rimon B. and Freund H. R. (1992). *Candida* sepsis during total parenteral nutrition: An endogenous infection indicating the severity of patients' disease state. Clin Nutr. 11(4): 240–243.

Cashikar A. G., Duennwald M. and Lindquist S. L. (2005). A chaperone pathway in protein disaggregation. Hsp26 alters the nature of protein aggregates to facilitate reactivation by Hsp104. J Biol Chem. 280: 23869–23875.

Castelo Branco S., Ferreira A. T., Saraiva S., Silva M. J. and Garcia T. (2017). Fungal Gastroduodenitis. Eur. J. Case Rep. Intern. Med. p. 2.

Chandra J., Kuhn D. M., Mukherjee P. K., Hoyer L. L., McCormick T. and Ghannoum M. A. (2001). Biofilm formation by the fungal pathogen *Candida albicans*: development, architecture, and drug resistance. Journal of Bacteriology. 183: 5385–5394.

Chen G., Rubinstein B. and Li R. (2012). Whole Chromosome Aneuploidy: Big Mutations Drive Adaptation by Phenotypic Leap. BioEssays. 34: 893–900.

Chen H., Zhou X., Ren B. and Cheng L. (2020). The regulation of hyphae growth in *Candida albicans*. 11: 337–348.

Cheng G., Wozniak K., Wallig M. A., Fidel P. L. Jr., Trupin S. R. and Hoyer L. L. (2005). Comparison between *Candida albicans* agglutinin–like sequence gene expression patterns in human clinical specimens and models of vaginal candidiasis. Infect Immun. 73: 1656–63.

Cho I., Jackson M. R. and Swift J (2016). Roles of Cross–Membrane Transport and Signalling in the Maintenance of Cellular Homeostasis. Cell. Mol. 9: 234–246.

Citiulo F., Jacobsen I. D., Miramón P., Schild L., Brunke S. and Zipfel P. (2012). *Candida albicans* scavenges host zinc via Pra1 during endothelial invasion. PLoS Pathog. 8: e1002777.

Ciurea C. N., Kosovski I.–B., Mare A. D., Toma F., Pintea–Simon I. A. and Man A. (2020). *Candida* and Candidiasis—Opportunism versus Pathogenicity: A Review of the Virulence Traits. Microorganisms. 8: 857.

Clatworthy A. E., Pierson E. and Hung D. T. (2007). Targeting Virulence: A New Paradigm for Antimicrobial Therapy. Nat. Chem. Biol. 3: 541–548.

Cleary I. A., Reinhard S. M., Miller C. L., Murdoch C., Thornhill M. H. and Lazzell A. L. *Candida albicans* adhesin Als3p is dispensable for virulence in the mouse model of disseminated candidiasis. Microbiology. 157: 1806–1815.

Coleman J. J. and Mylonakis E. (2009). Efflux in Fungi: La Piece de resistance. PLoS Pathog. 5: e1000486.

Cordeiro R. A., Evangelista A. J. J., Serpa R., Marques F. J. F., Melo C. V. S., Oliveira J. S., Franco J. D. S., Alencar L. P., Bandeira T. and Brilhante R. S. N. (2016). Inhibition of Heat-Shock Protein 90 Enhances the Susceptibility to Antifungals and Reduces the Virulence of *Cryptococcus neoformans*/*Cryptococcus gattii* Species Complex. Microbiology. 162: 309–317.

Coronado–Castellote L. and Jiménez–Soriano Y. (2013). Clinical and microbiological diagnosis of oral candidiasis. J Clin Exp Dent. 5(5): e279–e286.

Correia A., Lermann U., Teixeira L., Cerca F., Botelho S. and da Costa R. M. (2010). Limited role of secreted aspartyl proteinases Sap1 to Sap6 in *Candida albicans* virulence and host immune response in murine haematogenously disseminated candidiasis. Infect Immun. 78: 4839–4849.

Costa–de–Oliveira S. and Rodrigues A. G. (2020). *Candida albicans* Antifungal Resistance and Tolerance in Bloodstream Infections: The Triad Yeast–Host–Antifungal. Microorganisms. 8: 154.

Costanzo M., Baryshnikova A., Bellay J., Kim Y., Spear E. D., Sevier C. S., Ding H., Koh J. L., Toufighi K. and Mostafavi S. (2010). The Genetic Landscape of a Cell. Science. 327: 425–431.

Costanzo M., Baryshnikova A., Myers C. L., Andrews B. and Boone C. (2011). Charting the Genetic Interaction Map of a Cell. Curr. Opin. Biotechnol. 22: 66–74.

Costanzo M., VanderSluis B., Koch E. N., Baryshnikova A., Pons C., Tan G., Wang W., Usaj M., Hanchard J. and Lee S. D. (2016). A Global Genetic Interaction Network Maps a Wiring Diagram of Cellular Function. Science. 353: aaf1420.

Coste A., Selmecki A., Forche A., Diogo D., Bougnoux M. E., d'Enfert C., Berman J. and Sanglard D. (2007). Genotypic Evolution of Azole Resistance Mechanisms in Sequential *Candida albicans* Isolates. Eukaryotic Cell. 6: 1889–1904.

Coste A., Turner V., Ischer F., Morschhauser J., Forche A., Selmecki A., Berman J., Bille J. and Sanglard D. (2006). A Mutation in Tac1p, A Transcription Factor Regulating CDR1 and CDR2, is Coupled with Loss of Heterozygosity at Chromosome 5 to Mediate Antifungal Resistance in *Candida albicans*. Genetics. 17: 2139–2156.

Cowen L. E., Singh S. D., Köhler J. R., Collins C., Zaas A. K. and Schell W. A. (2009). Harnessing Hsp90 function as a powerful, broadly effective therapeutic strategy for fungal infectious disease. 106: 2818–2823.

Cowen L. E., Singh S. D., Kohler J. R., Collins C., Zaas A. K., Schell W. A., Aziz H., Mylonakis E., Perfect J. R. and Whitesell L. (2009). Harnessing Hsp90 Function as a Powerful, Broadly Effective Therapeutic Strategy for Fungal Infectious Disease. Proc. Natl. Acad. Sci. USA. 106: 2818–2823.

Csank C., Schröppel K., Leberer E., Harcus D., Mohamed O. and Meloche S. Roles of the *Candida albicans* mitogen–activated protein kinase homolog, Cek1p, in hyphal development and systemic candidiasis. Infect Immun. 66: 2713–2721.

Davis D. A. (2009). How human pathogenic fungi sense and adapt to pH: the link to virulence. Curr Opin Microbiology. 12: 365–370.

Davis D., Edwards J. E. Jr., Mitchell A. P. and Ibrahim A. S. (2000). *Candida albicans* RIM101 pH response pathway is required for host–pathogen interactions. Infect Immun. 68: 5953–5959.

De Bernardis F., Mühlschlegel F. A., Cassone A. and Fonzi W. A. (1998). The pH of the host niche controls gene expression in and virulence of *Candida albicans*. Infect Immun. 66: 3317–3325.

Diezmann S., Michaut M., Shapiro R. S., Bader G. D. and Cowen L. E. (2012). Mapping the Hsp90 Genetic Interaction Network in *Candida albicans* Reveals Environmental Contingency and Rewired Circuitry. PLoS Genet. 8: e1002562.

Dunkel N., Blass J., Rogers P. D. and Morschhauser J. (2008). Mutations in the Multi–Drug Resistance Regulator MRR1, followed by Loss of Heterozygosity, are the Main Cause of MDR1 Overexpression in Fluconazole–Resistant *Candida albicans* Strains. Mol. Microbiology. 69: 827–840.

Dunkel N., Liu T. T., Barker K. S., Homayouni R., Morschhauser J. and Rogers P. D. (2008). A Gain–of–Function Mutation in the Transcription Factor Upc2p Causes Upregulation of Ergosterol Biosynthesis Genes and Increased Fluconazole Resistance in a Clinical *Candida albicans* Isolate. Eukaryotic Cell. 7: 1180–1190.

Edwards J. E. Jr., Schwartz M. M., Schmidt C. S., Sobel J. D., Nyirjesy P., Schodel F., Marchus E., Lizakowski M., DeMontigny E. A. and Hoeg J. (2018). A Fungal Immunotherapeutic Vaccine (NDV-3A) for Treatment of Recurrent Vulvovaginal Candidiasis-A Phase 2 Randomised, Double-Blind, Placebo-Controlled Trial. Clin. Infect. Dis. 66: 1928–1936.

Ellis D. (2002). Amphotericin B: Spectrum and Resistance. J. Antimicrob. Chemotherapy. 49: 7–10.

Ene I. V., Adya A. K., Wehmeier S., Brand A. C., MacCallum D. M. and Gow N. A. (2012). Host carbon sources modulate cell wall architecture, drug resistance and virulence in a fungal pathogen. Cell Microbiol. 14: 1319–1335.

Eschenauer G. A., Kwak E. J., Humar A., Potoski B. A., Clarke L. G., Shields R. K., Abdel–Massih R., Silveira F. P., Vergidis P. and Clancy C. J. (2015). Targeted Versus Universal Antifungal Prophylaxis Among Liver Transplant Recipients. Am. J. Transplant. 15: 180–189.

Espinosa–Hernández V. M., Morales–Pineda V. and ótínez–Herrera E. (2020). Skin infections caused by emerging *Candida* species. Current Fungal Infection Reports. 14: 99–105

Eyles S. J. and Gierasch L. M. (2010). Nature's molecular sponges: small heat shock proteins grow into their chaperone roles. 107: 2727–2728.

Faith J. J., Guruge J. L., Charbonneau M., Subramanian S., Seedorf H., Goodman A. L., Clemente J. C., Knight R., Heath A. C. and Leibel R. L. (2013). The long–term stability of the human gut microbiota. Science. 341: 1237439.

Fangtham M., Magder L. S. and Petri M. A. (2014). Oral candidiasis in systemic lupus erythematosus. Lupus. 23(7): 684–690.

Fanning S. and Mitchell A. P. (2011). Fungal biofilms. PLoS Pathog. 8: e1002585.

Fidel P. L. Jr. (2004). History and new insights into host defence against vaginal candidiasis. Trends Microbiol. 12: 220–227.

Fidel P. L. Jr., Barousse M., Espinosa T., Ficarra M., Sturtevant J. and Martin D. H. (2004). An intravaginal live *Candida* challenge in humans leads to new hypotheses for the immunopathogenesis of vulvovaginal candidiasis. Infect Immun. 72: 2939–2946.

Finkel J. S. and Mitchell A. P. (2011). Genetic control of C*andida albicans* biofilm development. Nat Rev Microbiology. 9: 109–118.

Fiori A., Kucharíková S., Govaert G., Cammue B. P., Thevissen K. and Van Dijck P. (2012). The heat–induced molecular disaggregase Hsp104 of *Candida albicans* plays a role in biofilm formation and pathogenicity in a worm infection model. Eukaryot Cell. 11: 1012–1020.

Fleck C. B, Schöbel F. and Brock M. (2011). Nutrient acquisition by pathogenic fungi: nutrient availability, pathway regulation, and differences in substrate utilisation. Int J Med Microbiology. 301: 400–407.

Flowers S. A., Barker K. S., Berkow E. L., Toner G., Chadwick S. G., Gygax S. E., Morschhauser J. and Rogers P. D. (2012). Gain–of–Function Mutations in UPC2 are a Frequent Cause of ERG11 Upregulation in Azole–Resistant Clinical Isolates of *Candida albicans*. Eukaryotic Cell. 11: 1289–1299.

Flowers S. A., Colon B., Whaley S. G., Schuler M. A. and Rogers P. D. (2015). Contribution of Clinically Derived Mutations in ERG11 to Azole Resistance in *Candida albicans*. Antimicrob. Agents Chemotherapy. 59: 450–460.

Fonzi W. A. (1999). PHR1 and PHR2 of *Candida albicans* encode putative glycosidases required for proper crosslinking of β–1,3– and β–1,6–glucans. J Bacteriol. 181: 7070–7079.

Forche A., Abbey D., Pisithkul T., Weinzierl M. A., Ringstrom T., Bruck D., Petersen K. and Berman J. (2011). Stress Alters Rates and Types of Loss of Heterozygosity in *Candida albicans*. mBio. 2: e00129–11.

Fortún J., Meije Y., Buitrago M. J., Gago S., Bernal–Martinez L., Pemán J., Pérez M., Gómez–G Pedrosa E., Madrid N. and Pintado V. (2014). Clinical validation of a multiplex real–time PCR assay for detection of invasive candidiasis in intensive care unit patients. J. Antimicrob. Chemother. 69: 3134–3141.

Foxman B., Marsh J. V., Gillespie B. and Sobel J. D. (1998). Frequency and response to vaginal symptoms among white and African American women: results of a random digit dialling survey. J Womens Health. 7: 1167– 1174.

Fraimow H. S.and Abrutyn E. (1995). Pathogens resistant to antimicrobial agents. Epidemiology, molecular mechanisms, and clinical management. Infect. Dis. Clin. N. Am. 9: 497–530.

Frohner I. E., Bourgeois C., Yatsyk K., Majer O. and Kuchler K. (2009). *Candida albicans* cell surface superoxide dismutases degrade host–derived reactive oxygen species to escape innate immune surveillance. Mol Microbiology. 71:240–252.

Fu M. S., De Sordi L. and Mühlschlegel F. A. (2012). Functional characterisation of the small heat shock protein Hsp12p from *Candida albicans*. PLoS One. 7: e42894.

Fu Y., Ibrahim A. S., Fonzi W., Zhou X., Ramos C. F. and Ghannoum M. A. (1997). Cloning and characterisation of a gene (LIP1) which encodes a lipase from the pathogenic yeast *Candida albicans*. Microbiology. 143: 331–340.

Gabaldón T., Naranjo–Ortíz M. A. and Marcet–Houben M. (2016). Evolutionary genomics of yeast pathogens in the *Saccharomycotina*. FEMS Yeast Res. 16: fow064.

Gamaletsou M. N., Kontoyiannis D. P., Sipsas N. V., Moriyama B., Alexander E., Roilides E., Brause B. and Walsh T. J. (2012). *Candida* osteomyelitis: Analysis of 207 paediatric and adult cases (1970–2011). Clin. Infect. Dis. 55: 1338–1351.

Garcia M. C., Lee J. T., Ramsook C. B., Alsteens D., Dufrêne Y. F. and Lipke P. N. (2011). A role for amyloid in cell aggregation and biofilm formation. PLoS One. 6: e17632.

Garcia–Effron G., Katiyar S. K., Park S., Edlind T. D. and Perlin D. S. (2008). A Naturally Occurring Proline-To-Alanine Amino Acid Change in Fks1p in *Candida parapsilosis*, *Candida orthopsilosis*, and *Candida metapsilosis* Accounts for Reduced Echinocandin Susceptibility. Antimicrob. Agents Chemotherapy. 52: 2305–2312.

Garcia–Effron G., Lee S., Park S., Cleary J. D. and Perlin D. S. (2009). Effect of *Candida glabrata* FKS1 and FKS2 Mutations on Echinocandin Sensitivity and Kinetics of 1,3– β –D–Glucan Synthase: Implication for the Existing Susceptibility Breakpoint. Antimicrob. Agents Chemotherapy. 53: 3690–3699.

Garcia–Effron G., Park S. and Perlin D. S. (2009). Correlating Echinocandin MIC and Kinetic Inhibition of Fks1Mutant Glucan Synthases for *Candida albicans*: Implications for Interpretive Breakpoints. Antimicrob. Agents Chemotherapy. 53: 112–122.

Garcia–Rubio R., de Oliveira H. C., Rivera J. and Trevijano–Contador N. (2020). The Fungal Cell Wall: *Candida*, *Cryptococcus*, and *Aspergillus* Species. Front. Microbiol. 10: 2993.

Gaur M., Puri N., Manoharlal R., Rai V., Mukhopadhayay G., Choudhury D. and Prasad R. (2008). MFS Transportome of the Human Pathogenic Yeast *Candida albicans*. BMC Genomics. 9: 579.

Gauwerky K., Borelli C. and Korting H. C. (2009). Targeting Virulence: A New Paradigm for Antifungals. Drug Discovery Today. 14: 214–222.

Ghosh S., Navarathna D. H., Roberts D. D., Cooper J. T., Atkin A. L. and Petro T. M. (2009). Arginine–induced germ tube formation in *Candida albicans* is essential for escape from murine macrophage line RAä 264.7. Infect Immun. 77: 1596–1605.

Gill–Bona A., Amador–García A., Gil C. ad Monteoliva L. (2018). The external face of *Candida albicans*: A proteomic view of the cell surface and the extracellular environment. J. Proteom. 180: 70–79.

Gonzalez–Lara M. F. and Ostrosky–Zeichner L. (2020). Invasive Candidiasis. Semin. Respir. Crit. Care Med. 41: 3–12.

Gow N. A. and Hube B. (2012). Importance of the *Candida albicans* cell wall during commensalism and infection. Current Opinion in Microbiology. 15: 406–412.

Gudlaugsson O., Gillespie S., Lee K., Vande Berg J., Hu J., Messer S., Herwaldt L., Pfaller M. and Diekema D. (2003). Attributable mortality of nosocomial candidemia, revisited. Clinical Infectious Diseases. 37: 1172–1177.

Hall R. A. (2015). Dressed to impress: Impact of environmental adaptation on the *Candida albicans* cell wall. Mol. Microbiol. 97: 7–17.

Hall R. A., Turner K. J., Chaloupka J., Cottier F., De Sordi L. and Sanglard D. (2011). The quorum–sensing molecules farnesol/homoserine lactone and dodecanol operate via distinct modes of action in *Candida albicans*. Eukaryot Cell. 10: 1034–42.

Haron E., Vartivarian S., Anaissie E., Dekmezian R. and Bode, G. (1993). Primary *Candida* pneumonia. Experience at a large cancer centre and review of the literature. Medicine. 72: 137–142.

Harrison B. D., Hashemi J., Bibi M., Pulver R., Bavli D., Nahmias Y., Wellington M., Sapiro G. and Berman J. (2014). A Tetraploid Intermediate Precedes Aneuploid Formation in Yeasts Exposed to Fluconazole. PLoS Biol. 12: e1001815.

Haslbeck M., Walke S., Stromer T., Ehrnsperger M., White H. E. (1999). Hsp26: a temperature regulated chaperone. EMBO J. 18: 6744–6751.

Heilmann C. J., Schneider S., Barker K. S., Rogers P. D. and Morschhauser J. (2010). An A643T Mutation in the Transcription Factor Upc2p Causes Constitutive ERG11 Upregulation and Increased Fluconazole Resistance in *Candida albicans*. Antimicrob. Agents Chemotherapy. 54: 353–359.

Heymann P., Gerads M., Schaller M., Dromer F., Winkelmann G. and Ernst J. F. (2002). The siderophore iron transporter of *Candida albicans* (Sit1p/Arn1p) mediates uptake of ferrichrome–type siderophores and is required for epithelial invasion. Infect Immun. 70: 5246–5255.

Hill J. A., Ammar R., Torti D., Nislow C. and Cowen L. E. (2013). Genetic and Genomic Architecture of the Evolution of Resistance to Antifungal Drug Combinations. PLoS Genet. 9: e1003390.

Hirakawa M. P., Chyou D. E., Huang D., Slan A. R. and Bennett R. J. (2017). Parasex Generates Phenotypic Diversity de Novo and Impacts Drug Resistance and Virulence in *Candida albicans*. Genetics. 207: 1195–1211.

Höfs S., Mogavero S. and Hube B. (2016). Interaction of *Candida albicans* with host cells: Virulence factors, host defence, escape strategies and the microbiota. J. Microbiol. 54: 149–169.

Hood M. I. and Skaar E. P. (2012). Nutritional immunity: transition metals at the pathogen–host interface. Nat Rev Microbiology. 10: 525–537.

Hoot S. J., Smith A. R., Brown R. P. and White T. C. (2011). An A643V Amino Acid Substitution in Upc2p Contributes to Azole Resistance in Well–Characterised Clinical Isolates of *Candida albicans*. Antimicrob. Agents Chemother. 55: 940–942.

Hornby J. M., Jensen E. C., Lisec A. D., Tasto J. J., Jahnke B. and Shoemaker R. (2001). Quorum sensing in the dimorphic fungus *Candida albicans* is mediated by farnesol. Appl Environ Microbiology. 67: 2982–92.

Hoversten P., Kamboj A. K. and Katzka D. A. (2018). Infections of the esophagus: An update on risk factors, diagnosis and management. Dis. Oesophagus. p. 31.

Hoversten P., Otaki F. and Katzka D. A. (2019). Course of Esophageal Candidiasis and Outcomes of Patients at a Single Center. Clin. Gastroenterol. Hepatol. 17: 200–202.

Hsu J. F., Lai M. Y., Lee C. W., Chu S. M., Wu I. H., Huang H. R., Lee I. T., Chiang M. C., Fu R. H. and Tsai M. H. (2018). Comparison of the incidence, clinical features and outcomes of invasive candidiasis in children and neonates. BMC Infect. Dis. 18: 194. [Online] Available from: https://www.cdc.gov/fungal/diseases/candidiasis/thrush/index.html [Accessed on 30 January 2024]

Hube B., Sanglard D., Odds F. C., Hess D., Monod M. and Schäfer W. (1997). Disruption of each of the secreted aspartyl proteinase genes SAP1, SAP2, and SAP3 of *Candida albicans* attenuates virulence. Infect Immun. 65: 3529–3538.

Hube B., Stehr F., Bossenz M., Mazur A., Kretschmar M. and Schäfer W. (2000). Secreted lipases of *Candida albicans*: cloning, characterisation and expression analysis of a new gene family with at least ten members. Arch Microbiology. 174: 362–374.

Hurley R. and De Louvois J. (1979). *Candida* vaginitis. Postgrad Med J. 55: 645–647.

Ibrahim A. S., Luo G., Gebremariam T., Lee H., Schmidt C. S., Hennessey J. P. Jr., French S. W., Yeaman M. R., Filler S. G. and Edwards J. E. Jr. (2013). NDV-3 Protects Mice from Vulvovaginal Candidiasis Through T- and B-Cell Immune Response. Vaccine. 31: 5549–5556.

Inglis D. O., Arnaud M. B., Binkley J., Shah P., Skrzypek M. S. and Wymore F. (2012). The *Candida* genome database incorporates multiple *Candida* species: Multispecies search and analysis tools with curated gene and protein information for *Candida albicans* and *Candida glabrata*. Nucleic Acids Res. 40: D667–D674. PMID: 22064862.

Jabra–Rizk M. A., Kong E. F., Tsui C., Nguyen M. H., Clancy C. J., Fidel P. L. and Noverr M. (2016). *Candida albicans* pathogenesis: Fitting within the host–microbe damage response framework. Infect. Immun. 84: 2724–2739.

Jacobsen I. D., Wilson D., Wächtler B., Brunke S., Naglik J. R. and Hube B. (2012). *Candida albicans* dimorphism as a therapeutic target. Expert Rev Anti Infect Ther. 10:85–93.

Jensen R. H., Astvad K. M., Silva L. V., Sanglard D., Jorgensen R., Nielsen K. F., Mathiasen E. G., Doroudian G., Perlin D. S. and Arendrup M. C. (2015). Stepwise Emergence of Azole, Echinocandin and Amphotericin B Multidrug Resistance in–Vivo in *Candida albicans* Orchestrated by Multiple Genetic Alterations. J. Antimicrob. Chemotherapy. 70: 2551–2555.

Jeziorek M., Frej–M ˛adrzak M. and Choroszy–Król I. (2019). The influence of diet on gastrointestinal *Candida spp*. colonisation and the susceptibility of *Candida spp*. to antifungal drugs.. 70: 195–200.

Johnson M. D. and Perfect J. R. (2001). Fungal infections of the bones and joints. Curr. Infect. Dis. Rep. 3: 450–460.

Jordá T. and Puig S. (2020). Regulation of Ergosterol Biosynthesis in *Saccharomyces cerevisiae*. Genes. 11: 795.

Kathiravan M. K., Salake A. B., Chothe A. S., Dudhe P. B., Watode R. P., Mukta M. S. and Gadhwe S. (2012). The biology and chemistry of antifungal agents: A review. Bioorg. Med. Chem. 20: 5678–5698.

Kauffman C. A., Vazquez J. A., Sobel J. D., Gallis H. A., McKinsey D. S., Karchmer A. W., Sugar A. M., Sharkey P. K., Wise G. J. and Mangi R. (2000). Prospective multicentre surveillance study of funguria in hospitalised patients. Clin. Infect. Dis. 30: 14–18.

Kim J. and Sudbery P. (2011). *Candida albicans*, a major human fungal pathogen. Journal of microbiology 49: 171–177.

aKoh A. Y., Köhler J. R., Coggshall K. T., Van Rooijen N. and Pier G. B. (2008). Mucosal damage and neutropenia are required for *Candida albicans* dissemination. PLoS Pathog. 4: e35.

Kontoyiannis D. P., Reddy B. T., Torres H. A., Luna M., Lewis R. E., Tarrand J., Bodey G. P. and Raad I. I. (2002). Pulmonary candidiasis in patients with cancer: An autopsy study. Clin. Infect. Dis. 34: 400–403.

Kornitzer D. (2019). Regulation of *Candida albicans* hyphal morphogenesis by endogenous signals. J. Fungi.

Kumamoto C. A. (2008). Molecular mechanisms of mechanosensing and their roles in fungal contact sensing. Nat Rev Microbiology. 6: 6676–73.

Kumamoto C. A. and Pierce J. V. (2011). Immunosensing during colonisation by *Candida albicans*: Does it take a village to colonise the intestine? Trends in Microbiology. 19: 263–267.

Kumamoto C. A. and Vinces M. D. (2005). Alternative *Candida albicans* lifestyles: growth on surfaces. Annual Review of Microbiology. 59: 113–133.

LaFayette S. L., Collins C., Zaas A. K., Schell W. A., Betancourt–Quiroz M. and Gunatilaka A. A. (2010). PKC signalling regulates drug resistance of the fungal pathogen *Candida albicans* via circuitry comprised of Mkc1, calcineurin, and Hsp90. PLoS Pathog. 6: e1001069. PMID: 20865172.

Lamb D. C., Kelly D. E., Schunck W. H., Shyadehi A. Z., Akhtar M., Lowe D. J., Baldwin B. C. and Kelly S. L. (1997). The Mutation T315A in *Candida albicans* Sterol 14α–Demethylase Causes Reduced Enzyme Activity and Fluconazole Resistance Through Reduced Affinity. J. Biol. Chem. 272: 5682–5688.

Leach M. D., Stead D. A., Argo E. and Brown A. J. (2011). Identification of sumoylation targets, combined with inactivation of SMT3, reveals the impact of sumoylation upon growth, morphology, and stress resistance in the pathogen *Candida albicans*. Mol Biol Cell. 22: 687–702.

Lermann U. andMorschhäuser J. Secreted aspartic proteases are not required for invasion of reconstituted human epithelia by *Candida albicans*. (2008). Microbiology. 154: 3281–3295.

Li X. S., Reddy M. S., Baev D. and Edgerton M. (2003). *Candida albicans* Ssa1/2p is the cell envelope binding protein for human salivary histatin 5. J Biol Chem. 278: 28553–28556.

Li X. S., Sun J. N., Okamoto–Shibayama K. and Edgerton M. (2006). *Candida albicans* cell wall Ssa proteins bind and facilitate import of salivary histatin–5 required for toxicity. J Biol Chem. 281: 22453–22463.

Lindquist S. (1986). The heat–shock response. Annu Rev Biochem. 55: 1151–1191.

Lindquist S. (1992). Heat–shock proteins and stress tolerance in microorganisms. Curr Opin Genet Dev. 2: 748–755.

Lockhart S. R., Etienne K. A., Vallabhaneni S., Farooqi J., Chowdhary A., Govender N. P., Colombo A. L., Calvo B., Cuomo C. A. and Desjardins C. A. (2017). Simultaneous emergence of multidrug–resistant *Candida auris* on 3 continents confirmed by whole–genome sequencing and epidemiological analyses. Clin. Infect. Dis. 64: 134–140.

Logan C., Martin–Loeches I. and Bicanic T. (2020). Invasive candidiasis in critical care: Challenges and future directions. Intensive Care Med. 46: 2001–2014.

Lorenz M. C. and Fink G. R. (2001). The glyoxylate cycle is required for fungal virulence. Nature. 412: 83–86.

Lorenz M. C., Bender J. A. and Fink G. R. (2004). Transcriptional response of *Candida albicans* upon internalisation by macrophages. Eukaryot Cell. 3: 1076–1087. PMID: 15470236.

MacPherson S., Akache B., Weber S., De Deken X., Raymond M. and Turcotte B. (2005). *Candida albicans* Zinc Cluster Protein Upc2p Confers Resistance to Antifungal Drugs and is an Activator of Ergosterol Biosynthetic Genes. Antimicrob. Agents Chemotherapy. 49: 1745–1752.

Mahajan B., Bagul N., Desai R., Reddy M., Mahajan A., Shete A., Risbud A., Mane A. (2015). Pseudomembranous Type of Oral Candidiasis is Associated with Decreased Salivary Flow Rate and Secretory Immunoglobulin–A Levels. Mycopathologia. 180(1–2): 75–80.

Mamtani S. S., Aljanabi N. M., Gupta Rauniyar R. P., Acharya A. and Malik B. He. (2020). *Candida* Endocarditis: A Review of the Pathogenesis, Morphology, Risk Factors, and Management of an Emerging and Serious Condition. Cureus. pp. 12.

Marichal P., Koymans L., Willemsens S., Bellens D., Verhasselt P., Luyten W., Borgers M., Ramaekers F. C., Odds F. C. and Vanden Bossche H. (1999). Contribution of Mutations in the Cytochrome P450 14–α–Demethylase (Erg11p, Cyp51p) to Azole Resistance in *Candida albicans*. Microbiology. 145: 2701–2713.

Maródi L. (2014). *Mucocutaneous candidiasis*. In: Stiehm's Immune deficiencies. Vol. 1. Tokyo Academic Press. p. 775–802.

Martel C. M., Parker J. E., Bader O., Weig M., Gross U., Warrilow A. G., Rolley N., Kelly D. E. and Kelly S. L. (2010). Identification and Characterisation of Four Azole–Resistant erg3 Mutants of *Candida albicans*. Antimicrob. Agents Chemotherapy. 54: 4527–4533.

Martel C. M., Parker J. E., Bader O., Weig M., Gross U., Warrilow A. G., Kelly D. E. and Kelly S. L. (2010). A Clinical Isolate of *Candida albicans* with Mutations in ERG11 (Encoding Sterol 14alpha–Demethylase) and ERG5 (Encoding C22 Desaturase) is Cross Resistant to Azoles and Amphotericin B. Antimicrob. Agents Chemotherapy. 54: 3578–3583.

Mavor A. L., Thewes S. and Hube B. (2005). Systemic fungal infections caused by *Candida* species: Epidemiology, infection process and virulence attributes. Curr Drug Targets. 6: 863–874.

Mayer F. L., Wilson D., Jacobsen I. D., Miramón P., Große K. and Hube B. (2012). The novel *Candida albicans* transporter Dur31 Is a multi–stage pathogenicity factor. PLoS Pathog. 8: e1002592.

Mayer F. L., Wilson D., Jacobsen I. D., Miramón P., Slesiona S. and Bohovych I. M. (2012). Small but crucial: the novel small heat shock protein Hsp21 mediates stress adaptation and virulence in *Candida albicans*. PLoS One. 7: e38584.

Miller D. J. and Mejicano G. C. (2001). Vertebral osteomyelitis due to *Candida* species: Case report and literature review. Clin. Infect. Dis. 33: 523–530.

Millsop J. W. and Fazel N. (2016). Oral candidiasis. Clin Dermatol. 34(4): 487–494.

Mohamed A. A., Lu X. L. and Mounmin F. A. (2019). Diagnosis and Treatment of Oesophageal Candidiasis: Current Updates. Can. J. Gastroenterol. Hepatol. p. 2019.

Mohammad D. S. (2014). On the role of genomic islands in bacterial pathogenicity and antimicrobial resistance. Cloud Publ. Int. J. Adv. Biotechnol. Bioeng. 2: 18–36.

Molero G., Díez–Orejas R., Navarro–García F., Monteoliva L., Pla J., Gil, C., Sánchez–Pérez, M. and Nombela C. (1998). *Candida albicans*: Genetics, dimorphism and pathogenicity. Int. Microbiol. 1: 95–106.

Mora C., Tittensor D. P., Adl S., Simpson A. G.B. and Worm B. (2011). How many species are there on Earth and in the ocean? PLoS Biol. 9: e1001127.

Morad H. O. J., Wild A.–M., Wiehr S., Davies G., Maurer A., Pichler B. J. and Thornton C.R. (2018). Pre–clinical Imaging of Invasive Candidiasis Using Immuno-PET/MR. Front. Microbiol. 9: 1996.

Moran G. P., Coleman D. C. and Sullivan D. J. (2012). *Candida albicans* versus *Candida dubliniensis*: Why Is *C. albicans* More Pathogenic? Int J Microbiol. 2012: 205921.

Morio F., Loge C., Besse B., Hennequin C. and Le Pape P. (2010). Screening for Amino Acid Substitutions in the *Candida albicans* Erg11 Protein of Azole–Susceptible and Azole–Resistant Clinical Isolates: New Substitutions and a Review of the Literature. Diagn. Microbiol. Infect. Dis. 66: 373–384.

Mount H. O., Revie N. M., Todd R. T., Anstett K., Collins C., Costanzo M., Boone C., Robbins N., Selmecki A. and Cowen L. E. (2018). Global Analysis of Genetic Circuitry and Adaptive Mechanisms Enabling Resistance to the Azole Antifungal Drugs. PLoS Genet. 14: e1007319.

Moyes D. L., Richardson J. P. and Naglik J. R. (2015). *Candida albicans*–epithelial interactions and pathogenicity mechanisms: Scratching the surface. Virulence. 6: 338–346.

Mühlschlegel F. A. and Fonzi W. A. (1997). PHR2 of *Candida albicans* encodes a functional homolog of the pH–regulated gene PHR1 with an inverted pattern of pH–dependent expression. Mol Cell Biol. 17: 5960–5967.

Mukaremera L., Lee K. K., Mora–Montes H.M. and Gow N. A. R. (2017). *Candida albicans* yeast, pseudohyphal and hyphal morphogenesis differentially affects immune recognition. Front. Immunol. 8: 1.

Murciano C., Moyes D. L., Runglall M., Tobouti P., Islam A. and Hoyer L. L. (2012). Evaluation of the role of *Candida albicans* agglutinin–like sequence (Als) proteins in human oral epithelial cell interactions. PLoS One. 7: e33362.

Mushi M. F., Ngeta N., Mirambo M. M. and Mshana S. E. (2018). Predictors of esophageal candidiasis among patients attending endoscopy unit in a tertiary hospital, Tanzania: A retrospective cross–sectional study. Afr. Health Sci. 18: 66–71.

Naglik J. R., Challacombe S. J. and Hube B. (2003). *Candida albicans* secreted aspartyl proteinases in virulence and pathogenesis. Microbiol Mol Biol Rev. 67: 400–428.

Naglik J. R., König A., Hube B. and Gaffen S. L. (2017). *Candida albicans*–epithelial interactions and induction of mucosal innate immunity. Curr. Opin. Microbiol. 40: 104–112.

Narberhaus F. (2002). Alpha–crystallin–type heat shock proteins: socializing minichaperones in the context of a multichaperone network. Microbiol Mol Biol Rev. 66:64–93.

Neofytos D., Fishman J. A. and Horn D. (2010). Epidemiology and outcome of invasive fungal infections in solid organ transplant recipients. Transplant Infectious Disease. 12: 220–229

Nguyen M. H., Wissel M. C., Shields R. K., Salomoni M. A., Hao, B., Press E. G., Shields R. M., Cheng S., Mitsani D. and Vadnerkar A. (2012). Performance of *Candida* real–time polymerase chain reaction, β–D–glucan assay and blood cultures in the diagnosis of invasive candidiasis. Clin. Infect. Dis. 54: 1240–1248.

Niewerth M. and Korting H. C. (2001). Phospholipases of *Candida albicans*. Mycoses. 44: 361–367. PMID: 11766099.

Niimi K., Monk B. C., Hirai A., Hatakenaka K., Umeyama T., Lamping E., Maki K., Tanabe K., Kamimura T. and Ikeda F. (2010). Clinically Significant Micafungin Resistance in *Candida albicans* Involves Modification of a Glucan Synthase Catalytic Subunit GSC1 (FKS1) Allele Followed by Loss of Heterozygosity. J. Antimicrob. Chemother. 65: 842–852.

Nobile C. J. and Johnson A. D. (2015). *Candida albicans* Biofilms and Human Disease. Annu. Rev. Microbiol. 69: 71–92.

Nobile C. J., Fox E. P., Nett J. E., Sorrells T. R., Mitrovich Q. M. and Hernday A. D. (2012). A recently evolved transcriptional network controls biofilm development in *Candida albicans*. Cell. 148: 126–138.

Nobile C. J., Nett J. E., Hernday A. D., Homann O. R., Deneault J. S. and Nantel A. (2009). Biofilm matrix regulation by *Candida albicans* Zap1. PLoS Biol. 7: e1000133.

Odds F. C. (1998). *Candida* and Candidosis. Second ed. Bailliere Tindall, London, United Kingdom. p. 21.

Odds F. C. and Bernaerts R. (1994). CHROMagar *Candida*, a new differential isolation medium for presumptive identification of clinically important *Candida* species. J Clin Microbiol. 32(8): 1923–1929.

Odds F. C., Brown A. J. and Gow N. A. (2003). Antifungal agents: Mechanisms of action. Trends Microbiol. 11: 272–279.

Onyewu C., Blankenship J. R., Del Poeta M.and Heitman J. (2003). Ergosterol Biosynthesis Inhibitors Become Fungicidal when Combined with Calcineurin Inhibitors Against *Candida albicans*, *Candida glabrata* and *Candida krusei*. Antimicrob. Agents Chemother. 47: 956–964.

Pappas P. G., Kauffman C. A., Andes D. R., Clancy C. J., Marr K. A., Ostrosky–Zeichner L., Reboli A. C., Schuster M. G., Vazquez J. A. and Walsh T. J. (2015). Clinical Practice Guideline for the Management of Candidiasis: 2016 Update by the Infectious Diseases Society of America. Clin. Infect. 62: e1–e50.

Pappas P. G., Kauffman C. A., Andes D. R., Clancy C.J., Marr K. A., Ostrosky–Zeichner L., Reboli A. C., Schuster M. G., Vazquez J. A. and Walsh T. J. (2015). Clinical practice guideline for the management of candidiasis: 2016 update by the Infectious Diseases Society of America. Clin. Infect. Dis. 62: e1–e50.

Pappas P. G., Kauffman C. A., Andes D., Benjamin D. K. Jr., Calandra T. F. and Edwards J. E. Jr. (2009). Infectious Diseases Society of America. Clinical practice guidelines for the management of candidiasis: update by the Infectious Diseases Society of America. Clin Infect Dis. 48: 503–535.

Pappas P. G., Lionakis M. S., Arendrup M. C., Ostrosky–Zeichner L. and Kullberg B. J. (2018). Invasive candidiasis. Nat. Rev. Dis. Prim. 4: 1–20.

Pappas P. G., Rex J. H., Lee J., Hamill R.J., Larsen R. A., Powderly W. and Dismukes W. E. (2003). A prospective observational study of candidemia: epidemiology, therapy, and influences on mortality in hospitalized adult and paediatric patients. Clinical Infectious Diseases. 37: 634–643.

Park H., Myers C. L., Sheppard D. C., Phan Q. T., Sanchez A. A. and E Edwards J. (2005). Role of the fungal Ras–protein kinase–A pathway in governing epithelial cell interactions during oropharyngeal candidiasis. Cell Microbiology. 7: 499–510.

Park S., Kelly R., Kahn J. N., Robles J., Hsu M. J., Register E., Li W., Vyas V., Fan H. and Abruzzo G. (2005). Specific Substitutions in the Echinocandin Target Fks1p Account for Reduced Susceptibility of Rare Laboratory and Clinical *Candida* spp. Isolates. Antimicrob. Agents Chemotherapy. 49: 3264–3273.

Perfect J. R. (2017). The antifungal pipeline: A reality check. Nat. Rev. Drug Discov. 16: 603–616.

Perfect J. R. (2017). The Antifungal Pipeline: A Reality Check. Nat. Rev. Drug Discovery. 16: 603–616.

Perlin D. S. (2007). Resistance to Echinocandin–Class Antifungal Drugs. Drug Resist. Updates. 10: 121–130.

Perlroth J., Choi B. and Spellberg B. (2007). Nosocomial fungal infections: epidemiology, diagnosis, and treatment. Med Mycol. 45: 321–346.

Pfaller M. A. (2012). Antifungal drug resistance: Mechanisms, epidemiology and consequences for treatment. Am. J. Med. 125: S3–S13.

Pfaller M. A. and Diekema D. J. (2007). Epidemiology of invasive candidiasis: a persistent public health problem. Clin Microbiol Rev. 20: 133–163.

Pfaller M. A. and Diekema D. J. (2010). Epidemiology of invasive mycoses in North America. Crit Rev Microbiol. 36: 1–53.

Pfaller M. A., Andes D. R., Diekema D. J., Horn D. L., Reboli A. C., Rotstein C., Franks B., Azie N. E. (2014). Epidemiology and outcomes of invasive candidiasis due to non–*albicans* species of *Candida* in 2496 patients: Data from the Prospective Antifungal Therapy (PATH) registry 2004–2008. PLoS ONE 2014: 9.

Pfaller M. A., Messer S. A., Woosley L. N. and Jones R. N. (2013). Castanheira, M. Echinocandin and triazole antifungal susceptibility profiles for clinical opportunistic yeast and mold isolates collected from 2010 to 2011: Application of new CLSI clinical breakpoints and epidemiological cut off values for characterisation of geographic. J. Clin. Microbiol. 51: 2571–2581.

Pietrella D., Rachini A., Torosantucci A., Chiani P., Brown A. J., Bistoni F., Costantino P., Mosci P., d'Enfert C. and Rappuoli R. (2010). A β-Glucan-Conjugate Vaccine and Anti-β-Glucan Antibodies Are Effective Against Murine Vaginal Candidiasis as Assessed by a Novel In-Vivo Imaging Technique. Vaccine. 28: 1717–1725.

Pittet D., Monod M., Suter P. M., Frenk E. and Auckenthaler R. (1994). *Candida* colonisation and subsequent infections in critically ill surgical patients. Annals of Surgery. 220: 751–758.

Revie N. M., Iyer K. R., Robbins N. and Cowen L. E. (2018). Antifungal Drug Resistance: Evolution, Mechanisms and Impact. Curr. Opin. Microbio. 45: 70–76.

Revie N. M., Iyer K. R., Robbins N., Cowen L. E. (2018). Antifungal Drug Resistance: Evolution, Mechanisms and Impact. Curr. Opin. Microbiol. 45: 70–76.

Reyna–Beltrán E., Isaac Bazán Méndez C., Iranzo M., Mormeneo S. and Pedro Luna–Arias J. (2019). *The Cell Wall of Candida albicans*: *A Proteomics View*. In: *Candida Albicans.* Sandai D. (Ed.). IntechOpen: London, UK.

Rezaei N., Moin M. and Pourpak Z. (2007). The clinical, immunohaematological and molecular study of Iranian patients with severe congenital neutropenia. Journal of Clinical Immunology. 27: 525–533

Richardson J. P., Ho J. and Naglik J. R. (2018). *Candida*–epithelial interactions. J. Fungi. 4: 22.

Richter K., Haslbeck M. and Buchner J. (2010). The heat shock response: Life on the verge of death. Mol Cell. 40: 253–266.

Robbins N., Caplan T. and Cowen L. E. (2017). Molecular Evolution of Antifungal Drug Resistance. Annu. Rev. Microbio. 71: 753–775.

Robbins N., Uppuluri P., Nett J., Rajendran R., Ramage G. and Lopez–Ribot J. L. (2011). Hsp90 governs dispersion and drug resistance of fungal biofilms. PLoS Pathog. 7: e1002257.

Robbins N., Wright G. D. and Cowen L. E. (2016). Antifungal Drugs: The Current Armamentarium and Development of New Agents. Microbiol. Spectrum. 4: No. FUNK–0002–2016.

Robertson K. D., Nagra N. and Mehta D. (2013). Oesophageal Candidiasis. StatPearls. 1: 64–65.

Rodrigues C., Rodrigues M., Silva S. and Henriques M. (2017). *Candida glabrata* biofilms: How far have we come? J. Fungi. 3: 11.

Rosati D., Bruno M., Jaeger M., Ten Oever J. and Netea M. G. (2020). Recurrent Vulvovaginal Candidiasis: An Immunological Perspective. Microorganisms. p. 8.

Rosołowski M. and Kierzkiewicz M. (2013). Aetiology, diagnosis and treatment of infectious esophagitis. Prz. Gastroenterol. 8: 333–337.

Ruhnke M. (2002). Skin and mucous membrane infections. In *Candida* and Candidiasis, Calderone R. A. (Ed.). ASM Press, Washington, DC. p. 307–325.

Sanglard D., Ischer F., Parkinson T., Falconer D. and Bille J. (2003). *Candida albicans* Mutations in the Ergosterol Biosynthetic Pathway and Resistance to Several Antifungal Agents. Antimicrob. Agents Chemotherapy. 47: 2404–2412.

Sardi J. C. O., Scorzoni L., Bernardi T., Fusco–Almeida A. M., Mendes Giannini M. J. S. (2013). *Candida* species: Current epidemiology, pathogenicity, biofilm formation, natural antifungal products and new therapeutic options. J. Med. Microbiol. 62: 10–24.

Schaller M., Korting H. C., Schäfer W., Bastert J., Chen W. and Hube B. (1999). Secreted aspartic proteinase (Sap) activity contributes to tissue damage in a model of human oral candidosis. Mol Microbiology. 34: 169–180.

Schmidt C. S., White C. J., Ibrahim A. S., Filler S. G., Fu Y., Yeaman M. R., Edwards J. E. Jr. and Hennessey J. P. Jr. (2012). NDV-3, a Recombinant Alum-Adjuvanted Vaccine for *Candida* and *Staphylococcus aureus*, is Safe and Immunogenic in Healthy Adults. Vaccine. 30: 7594–7600.

Schnabel R. M., Linssen C. F., Guion N., Van Mook W. N. and Bergmans D. C. (2014). *Candida* pneumonia in intensive care unit? Open Forum Infect. Dis. p. 1.

Segal B. H., Herbrecht R. and Stevens D. A. (2008). Defining responses to therapy and study outcomes in clinical trials of invasive fungal diseases: Mycoses study group and European Organisation for Research and Treatment of Cancer consensus criteria. Clinical Infectious Diseases. 47: 674–683

Sekirov I., Russell S. L., Caetano M., Antunes L. and Finlay B. B. (2010). Gut microbiota in health and disease. Physiol. Rev. 90: 859–904.

Selmecki A. M., Dulmage K., Cowen L. E., Anderson J. B. and Berman J. (2009). Acquisition of Aneuploidy Provides Increased Fitness During the Evolution of Antifungal Drug Resistance. PLoS Genet. 5: e1000705.

Selmecki A., Forche A. and Berman J. (2006). Aneuploidy and Isochromosome Formation in Drug-Resistant. Science. 313: 367–370.

Selmecki A., Forche A. and Berman J. (2010). Genomic Plasticity of the Human Fungal Pathogen *Candida albicans*. Eukaryotic Cell. 9: 991–1008.

Selmecki A., Gerami-Nejad M., Paulson C., Forche A. and Berman J. (2008). An Isochromosome Confers Drug Resistance In-Vivo by Amplification of Two Genes, ERG11 and TAC1. Mol. Microbiol. 68: 624–641.

Shapiro R. S., Uppuluri P., Zaas A. K., Collins C., Senn H. and Perfect J. R. (2009). Hsp90 orchestrates temperature–dependent *Candida albicans* morphogenesis via Ras1–PKA signaling. Curr Biol. 19: 621– 629.

Shapiro R. S., Uppuluri P., Zaas A. K., Collins C., Senn H., Perfect J. R., Heitman J. and Cowen L. E. (2009). Hsp90 Orchestrates Temperature-Dependent *Candida albicans* Morphogenesis via Ras1-PKA Signaling. Curr. Biol. 19: 621–629.

Sheehan D. J., Hitchcock C. A. and Sibley C. M. (1999). Current and emerging azole antifungal agents. Clin. Microbiol. Rev. 12: 40–79.

Shoham S., Nucci M. and Walsh T. J. (2011). *Mucocutaneous and Deeply Invasive Candidiasis*. In: Tropical Infectious Diseases. Guerrant R. L., Walker D. H. and Weller P. F. o'meara(Eds.). Elsevier Inc. Boston, MA, USA. p. 589–596. ISBN 9780702039355.

Siddharthan T., Karakousis P. C. and Checkley W. (2016). Empirical antifungal therapy in critically ill patients with sepsis. JAMA J. Am. Med. Assoc. 316: 1549–1550.

Silver P. M., Oliver B. G. and White T. C. (2004). Role of *Candida albicans* Transcription Factor Upc2p in Drug Resistance and Sterol Metabolism. Eukaryotic Cell. 3: 1391–1397.

Singh S. D., Robbins N., Zaas A. K., Schell W. A., Perfect J. R. and Cowen L. E. (2009). Hsp90 governs echinocandin resistance in the pathogenic yeast *Candida albicans* via calcineurin. PLoS Pathog. 5: e1000532.

Singh S. D., Robbins N., Zaas A. K., Schell W. A., Perfect J. R. and Cowen L. E. (2009). Hsp90 Governs Echinocandin Resistance in the Pathogenic Yeast *Candida albicans* via Calcineurin. PLoS Pathog. 5: e1000532.

Singh S., Uppuluri P., Mamouei Z., Alqarihi A., Elhassan H., French S., Lockhart S. R., Chiller T., Edwards J. E. Jr. and Ibrahim A. S. (2019). The NDV-3A Vaccine Protects Mice from Multidrug Resistant *Candida auris* Infection. PLoS Pathog. 15: e1007460.

Sivabalan S., Mahadevan S. and Srinath M. V. (2014). Recurrent oral thrush. Indian J Pediatr. 81(4): 394–396.

Sobel J. D. (1985). Epidemiology and pathogenesis of recurrent vulvovaginal candidiasis. Am J Obstet Gynecol. 152: 924–935.

Sobel J. D. (1992). Pathogenesis and treatment of recurrent vulvovaginal candidiasis. Clin Infect Dis. 14: 48–53.

Sobel J. D. (1997). Vaginitis. N Engl J Med. 337(26): 1896–1903.

Sobel J. D. (2007). Vulvovaginal candidosis. Lancet. 369: 1961–1971.

Sobel J. D., Faro S., Force R. W., Foxman B., Ledger W. J. and Nyirjesy P. R. (2003). Vulvovaginal candidiasis: epidemiologic, diagnostic and therapeutic considerations. Am J Obstet Gynecol. 178(2): 203–211.

Soll D. R. (2009). Why does *Candida albicans* switch? FEMS Yeast Res. 9: 973–89.

Soloviev D. A., Jawhara S. and Fonzi W. A. (2011). Regulation of innate immune response to *Candida albicans* infections by αMβ2–Pra1p interaction. Infect Immun. 79: 1546–1558.

Specht C. A., Lee C. K., Huang H., Tipper D. J., Shen Z. T., Lodge J. K., Leszyk J., Ostroff G. R. and Levitz S. M. (2015). Protection Against Experimental Cryptococcosis Following Vaccination with Glucan Particles Containing *Cryptococcus* Alkaline Extracts. MBio. 6: e01905– e01915.

Spellberg B. J., Ibrahim A. S., Avanesian V., Fu Y., Myers C., Phan Q. T., Filler S. G., Yeaman M. R. and Edwards J. E. Jr. (2006). Efficacy of the Anti-*Candida* rAls3p-N or rAls1p-N Vaccines Against Disseminated and Mucosal Candidiasis. J. Infect. Dis. 194: 256–260.

Spellberg B., Marr K. and Filler S. G. (2012). *Candida*: What Should Clinicians and Scientists Be Talking About? In *Candida and* Candidiasis, Calderone R. A. and Clancy C. J. (Ed.). ASM Press, Washington, DC. p. 225–242.

Spitzer M., Robbins N. and Wright G. D. (2017). Combinatorial Strategies for Combating Invasive Fungal Infections. Virulence. 8: 169–185.

Staab J. F., Bradway S. D., Fidel P. L. and Sundstrom P. (1999). Adhesive and mammalian transglutaminase substrate properties of *Candida albicans* Hwp1. Science. 283: 1535–1538.

Staib P and Morschhäuser J. (2007). Chlamydospore formation in *Candida albicans* and *Candida dubliniensis*–an enigmatic developmental programme. Mycoses. 50: 1–12.

Stamatiades G. A., Ioannou P., Petrikkos G. and Tsioutis C. (2018). Fungal infections in patients with inflammatory bowel disease: A systematic review. Mycoses. 61: 366–376.

Sucher A. J., Chahine E. B. and Balcer H. E. (2009). Echinocandins: The newest class of antifungals. Ann. Pharmacother. 43: 1647–1657.

Sudbery P. E. (2011). Growth of *Candida albicans* hyphae. Nat Rev Microbiol 2011. 9: 737–48.

Sudbery P., Gow N. and Berman J. 2004). The distinct morphogenic states of *Candida albicans*. Trends Microbiology. 12: 317–24.

Sui X., Yan L. and Jiang Y. Y. (2017). The Vaccines and Antibodies Associated with Als3p for Treatment of *Candida albicans* Infections. Vaccine. 35: 5786–5793.

Sun J. N., Li W., Jang W. S., Nayyar N., Sutton M. D. and Edgerton M. (2008). Uptake of the antifungal cationic peptide Histatin 5 by *Candida albicans* Ssa2p requires binding to non–conventional sites within the ATPase domain. Mol Microbiology. 70: 1246–1260.

Sun J. N., Solis N. V., Phan Q. T., Bajwa J. S., Kashleva H. and Thompson A. (2010). Host cell invasion and virulence mediated by *Candida albicans* Ssa1. PLoS Pathog. 6: e1001181.

Sundstrom P. (2002). Adhesion in *Candida* spp. Cell Microbiology. 4: 461–9.

Sundstrom P., Balish E. and Allen C. (2002). Essential role of the *Candida albicans*. transglutaminase substrate, hyphal wall protein 1, in lethal oesophageal candidiasis in immunodeficient mice. J Infect Dis. 185: 521–530.

Taff H. T., Nett J. E., Zarnowski R., Ross K. M., Sanchez H. and Cain M. T. (2012). A *Candida* biofilm–induced pathway for matrix glucan delivery: Implications for drug resistance. PLoS Pathog. 8: e1002848.

Taylor B. N., Hannemann H., Sehnal M., Biesemeier A., Schweizer A. and Röllinghoff M. (2005). Induction of SAP7 correlates with virulence in an intravenous infection model of candidiasis but not in a vaginal infection model in mice. Infect Immun. 73: 7061–7063.

Tong Y. and Tang J. (2017). *Candida albicans* infection and intestinal immunity. Microbiol. Res. 198: 27–35.

Torosantucci A., Bromuro C., Chiani P., De Bernardis F., Berti F., Galli C., Norelli F., Bellucci C., Polonelli L. and Costantino P. (2005). A Novel Glyco-Conjugate Vaccine Against Fungal Pathogens. J. Exp. Med. 202: 597–606.

Tsao S., Rahkhoodaee F. and Raymond M. (2009). Relative Contributions of the *Candida albicans* ABC Transporters Cdr1p and Cdr2p to Clinical Azole Resistance. Antimicrob. Agents Chemotherapy. 53: 1344–1352.

Uppuluri P., Chaturvedi A. K., Srinivasan A., Banerjee M., Ramasubramaniam A. K. and Köhler J. R. (2010). Dispersion as an important step in the *Candida albicans* biofilm developmental cycle. PLoS Pathog. 6: e1000828.

Vasicek E. M., Berkow E. L., Flowers S. A., Barker K. S. and Rogers P. D. (2014). UPC2 is Universally Essential for Azole Antifungal Resistance in *Candida albicans*. Eukaryotic Cell. 13: 933–946.

Vazquez J. A., Arganoza M. T., Boikov D., Yoon S., Sobel J. D. and Akins R. A. (1998). Stable Phenotypic Resistance of *Candida* Species to Amphotericin B Conferred by Preexposure to Subinhibitory Levels of Azoles. J. Clin. Microbiology. 36: 2690–2695.

Verstrepen K. J. and Klis F. M. (2006). Flocculation, adhesion and biofilm formation in yeasts. Mol Microbiology. 60: 5–15.

Viana R., Dias O., Lagoa D., Galocha M., Rocha I. and Teixeira M. C. (2020). Genome–Scale Metabolic Model of the Human Pathogen *Candida albicans*: A Promising Platform for Drug Target Prediction. J. Fungi. 6: 171.

Vylkova S., Carman A. J., Danhof H. A., Collette J. R., Zhou H. and Lorenz M. C. (2011). The fungal pathogen *Candida albicans* auto induces hyphal morphogenesis by raising extracellular pH. MBio. 2: e00055–11.

Wächtler B., Citiulo F., Jablonowski N., Förster S., Dalle F. and Schaller M. (2012). *Candida albican*s–epithelial interactions: dissecting the roles of active penetration, induced endocytosis and host factors on the infection process. PLoS One. 7: e36952.

Walker G. M. and White N. A. (2017). *Introduction to Fungal Physiology*. In: Fungi: Biology and Applications, Kavanagh K. (Ed.). WileyBlackwell: Chichester, UK. p. 4. ISBN 978–1–119–37432–37439.

Weete J. D., Abril M. and Blackwell M. (2010). Phylogenetic distribution of fungal sterols. PLoS One. p. 5.

Weissman Z. and Kornitzer D. (2004). A family of *Candida* cell surface haem–binding proteins involved in haemin and haemoglobin–iron utilisation. Mol Microbiology. 53: 1209–1220.

Weitkamp J. H. and Nania J. J. (2007). *Infectious Diseases*. In: Neonatal Neurology. Fenichel G. M. (Ed.). Elsevier Inc. Boston, MA, USA. pp. 109–141. ISBN 9780443067242.

Whaley S. G., Caudle K. E., Vermitsky J. P., Chadwick S. G., Toner G., Barker K. S., Gygax S. E. and Rogers P. D. (2014). UPC2A is Required for High–Level Azole Antifungal Resistance in *Candida glabrata*. Antimicrob. Agents Chemotherapy. 58: 4543–4554.

Wiederhold N. P. (2017). Antifungal resistance: Current trends and future strategies to combat. Infect. Drug Resist. 10: 249–259.

Wisplinghoff H., Seifert H., Tallent S. M., Bischoff T., Wenzel R. P. and Edmond M.B. (2003). Nosocomial bloodstream infections in paediatric patients in United States hospitals: epidemiology, clinical features and susceptibilities. The Paediatric Infectious Disease Journal. 22: 686–691.

Wu Y., Gao N., Li C., Gao J. and Ying C. (2017). A Newly Identified Amino Acid Substitution T123I in the 14α–Demethylase (Erg11p) Of *Candida albicans* Confers Azole Resistance. FEMS Yeast Res. 17: fox012.

Wysong D. R., Christin L., Sugar A. M., Robbins P. W. and Diamond R. D. (1998). Cloning and sequencing of a *Candida albicans* catalase gene and effects of disruption of this gene. Infect Immun. 66: 1953–1961.

Xiang M. J., Liu J. Y., Ni P. H., Wang S., Shi C., Wei B., Ni Y. X. and Ge H. L. (2013). Erg11 Mutations Associated with Azole Resistance in Clinical Isolates of *Candida albicans*. FEMS Yeast Res. 386–393.

Xie Z., Thompson A., Sobue T., Kashleva H., Xu H. and Vasilakos J. (2012). *Candida albicans* biofilms do not trigger reactive oxygen species and evade neutrophil killing. J Infect Dis.

Yang F., Kravets A., Bethlendy G., Welle S. and Rustchenko E. (2013). Chromosome 5 Monosomy of *Candida albicans* Controls Susceptibility to Various Toxic Agents, Including Major Antifungals. Antimicrob. Agents Chemother. 57: 5026–5036.

Yang F., Teoh F., Tan A. S. M., Cao Y., Pavelka N. and Berman J. (2019). Aneuploidy Enables Cross-Adaptation to Unrelated Drugs. Mol. Biol. Evol. 36: 1768–1782.

Yapar N. (2014). Epidemiology and risk factors for invasive candidiasis. Ther. Clin. Risk Manag. 10: 95–105.

Zakikhany K., Naglik J. R., Schmidt–Westhausen A., Holland G., Schaller M. and Hube B. (2007). In vivo transcript profiling of *Candida albicans* identifies a gene essential for interepithelial dissemination. Cell Microbiology. 9: 2938–2954.

Zhu W. and Filler S. G. (2010). Interactions of *Candida albicans* with epithelial cells. Cell. Microbiol. 12: 273–282.

Zhu W., Filler S. G. (2010). Interactions of *Candida albicans* with epithelial cells. Cell Microbiology. 12: 273–82.

Zordan R. and Cormack B. (2012). *Adhesins on Opportunistic Fungal Pathogens*. In: *Candida* and Candidiasis, Calderone R. A. and Clancy C. J. (Eds.). ASM Press, Washington, DC. p. 243–259.

Anonymous. (2022). Tackling the emerging threat of antifungal resistance to human health. [online] Available from: https://www.nature.com/articles/s41579-022-00720-1 [Accessed 05 July 2024]

Bastos R. W., Rossato L., Goldman G. H., Santos D. A. (2021) Fungicide effects on human fungal pathogens: Cross-resistance to medical drugs and beyond. PLoS Pathog 17(12): e1010073

Calderone R. A. (2002). Introduction and historical perspectives. *Candida* and Candidiasis. p. 15-25. ASM Press. Washington, DC.

Chaieb K., Kouidhi B., Zmantar T., Mahdouani K. and Bakhrouf. (2011). Starvation survival of *Candida albicans* in various water microcosms. Journal of Basic Microbiology. 51 (4): 357 – 363.

Flowers S. A., Colón B., Whaley S. G., Schuler M. A. and Rogers P. D. (2015). Contribution of clinically derived mutations in ERG11 to azole resistance in *Candida albicans*. Antimicrob Agents Chemother 59: 450 –460.

Perry J. L. and Miller G. R. (1987). Umbelliferyl-labeled galactosaminide as an aid in identification of *Candida albicans*. 25: 2424-2425.

Popp C., Ramírez-Zavala B., Schwanfelder S., Krüger I. and Morschhäuser J. (2019). Evolution of fluconazole-resistant *Candida albicans* strains by drug-induced mating competence and parasexual recombination. 10 (1): e02740-18.

Ribeiro F. C., Rossoni R. D., de Barros P. P., Santos J. D., Fugisaki L. R. O., Leao M. P. V. and Junqueira J. C. (2019). Action mechanisms of probiotics on *Candida* spp. and candidiasis prevention: An update. p. 1.

Roudbarmohammadi S., Roudbary M., Bakshi B., Katiraee F., Mohammadi R. and Falahati M. (2016). ALS1 and ALS3 gene expression and biofilm formation in *Candida albicans* isolated from vuvlvovaginal candidiasis. 8 (5): 105.

Rouselle P., Freydiere A., Couillerot P., de Montclos H. and Gille Y. (1994). Rapid identification of *Candida albicans* by using Albicans ID and fluoroplate agar plates. 30: 3034-3036.

Yenisehirli G., Bulut N., Yenisehirli A. and Bulut Y. (2015). In Vitro Susceptibilities of *Candida albicans* isolates to Antifungal Agents in Tokat, Turkey. 8 (9): e28057.

Zangl I., Pap I-J., Aspöck C. and Schüller C. (2019). The role of *Lactobacillus* species in the control of *Candida* via biotrophic interactions. 7 (1): 3-4.

APPENDIX

Appendix I: Biochemical Assays for *Candida albicans*

Isolate	Germ Tube Test	Lactophenol Cotton Blue (LPCB) Assay	Chromogenic Agar Assay (CIA)
CA1	+	+	+
CA2	+	+	+
CA3	+	+	+
CA4	+	+	+
CA5	+	+	+
CA6	+	+	+
CA7	+	+	+
CA8	+	+	+
CA9	+	+	+
CA10	+	+	+
CA11	+	+	+
CA12	+	+	+
CA13	+	+	+
CA14	+	+	+
CA15	+	+	+
CA16	+	+	+
CA17	+	+	+
CA18	+	+	+
CA19	+	+	+
CA20	+	+	+

Key

+ Positive

Appendix II: Percentage Susceptibility, Resistance and Intermediate for *Candida albicans*

Characteristic	Fluconazole	Itriconazole	Posaconazole
Susceptible n (%)	0 (0%)	3 (15%)	10 (50%)
Intermediate n (%)	0 (0%)	0 (0%)	0 (0%)
Resistance n(%)	20 (100%)	17 (85%)	10 (50%)

Appendix III*:* Diameter Range (mm) for Antifungal Susceptibility, Intermediate and Resistance

Antifungal	Susceptibility	Intermediate	Resistance
Fluconazole	≥ 28mm	24 – 27mm	≤ 23mm
Itriconazole	≥ 27mm	23 – 26mm	≤ 22mm
Posaconazole	≥ 25mm	22 – 24mm	≤ 21mm

Appendix IV: Antibiogram for *Candida albicans* to Fluconazole, Itriconazole and Posaconazole

Isolate	Antifungal Susceptibility Test					
	Fluconazole	**Comment**	**Itriconazole**	**Comment**	**Posaconazole**	**Comment**
CA1	0 mm	R	0 mm	R	0 mm	R
CA2	0 mm	R	0 mm	R	30 mm	S
CA3	0 mm	R	10 mm	R	31 mm	S
CA4	0 mm	R	11 mm	R	30 mm	S
CA5	0 mm	R	8 mm	R	31 mm	S
CA6	8 mm	R	12 mm	R	28 mm	S
CA7	10 mm	R	0 mm	R	0 mm	R
CA8	0 mm	R	8 mm	R	0 mm	R
CA9	0 mm	R	9 mm	R	30 mm	S
CA10	11 mm	R	9 mm	R	16 mm	R
CA11	18 mm	R	12 mm	R	20 mm	R
CA12	0 mm	R	0 mm	R	0 mm	R
CA13	0 mm	R	10 mm	R	0 mm	R
CA14	9 mm	R	0 mm	R	26 mm	S
CA15	0 mm	R	0 mm	R	10 mm	R
CA16	0 mm	R	0 mm	R	0 mm	R
CA17	12 mm	R	29 mm	S	27 mm	S
CA18	0 mm	R	30 mm	S	28 mm	S
CA19	15 mm	R	27 mm	S	27 mm	S
CA20	0 mm	R	9 mm	R	9 mm	R

Key

R – Resistant

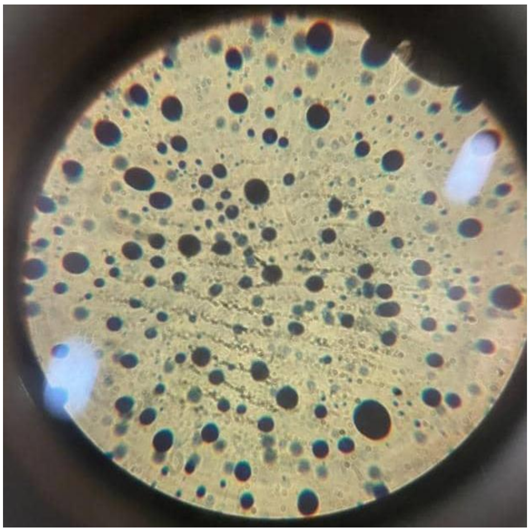
I – Intermediate

S – Susceptible

Appendix V: Positive Chromogenic Agar Assay for *Candida albicans* on Candida Ident Assay

Appendix VI: *Candida albicans* on Sabouraud Dextrose Agar

Appendeix VII: **Germ Tube Positive Assay for *Candida albicans***



Appendix VIII: **Lactophenol Cotton Blue Assay for *Candida albicans***